

PARIKSHA365 — STUDY NOTES

Physics — Common Book

SSC, RRB, State PSC, Banks (GA) — General Science

SSC

RRB

BANKS

STATE PSC

v 2026-04

The promise. Read this book end-to-end (with the usual two re-reads + active recall on the embedded prompts) and you will be able to attempt every quiz question in the Pariksha365 pool for this subject with a strong fundamental grasp.

Examiner's Blueprint — How Physics Gets Tested

EXAM ALERT

Physics questions follow a predictable pattern. They are rarely about solving equations — almost always about conceptual recall and application.

The examiner's favourite moves:

- **1. "Why does X happen?"** — pressure cooker, ice floating, sky colour. Pure understanding; every paper.
- **2. "Who discovered / invented X?"** — Röntgen, Faraday, Chadwick, Fleming. 1–2 per paper.
- **3. Vision defects** — myopia / hypermetropia and concave / convex lens. Never skipped.
- **4. Motor vs Generator** — which hand rule? A classic 1-mark gift.
- **5. Sound speed and echo distance** — straightforward calculation.
- **6. Alpha / Beta / Gamma radiation** — penetration, charge, what stops them.

Pure numerics (force calculations, wave equations) appear rarely. This book builds concepts first, then gives the numbers you must know cold.

Priority order for your study time:

1. Daily-life phenomena (Part X) — "why" questions; one in every paper
2. Instruments table (Part V) — 1-2 direct questions per paper
3. Discoverers and inventions (Part W) — constant presence
4. Sound: speed, echo, ranges (Part D) — easy marks
5. Light: mirror/lens types, vision defects, Rayleigh scattering (Part D)
6. Newton's laws with examples (Part A)
7. Electricity: Ohm's law, Indian supply, AC/DC (Part E)
8. Electromagnetism: motor/generator/transformer (Part E)
9. Nuclear: fission/fusion/radiation types (Part F)
10. Units, constants, formulas (Appendices)

How to read this book — your real study load

Total pages: 35. Don't let the page count scare you.

Mandatory reading — pages 1 to 25 (71% of the book)

That's the full syllabus. Master those 25 pages and you've covered every concept an examiner can fairly ask.

Bonus material — pages 26 to 35 (29% of the book)

Reference tables, compilations, and self-test material — useful in your last week of revision OR if you're aiming for a top rank.

Two study tracks

- **Just want to pass?** Read pages 1–25. Skim the appendices once before the exam.
 - **Want to top your batch?** Read the appendices too — they're how 70% becomes 90%.
-

Index — Table of Contents (clickable)

Pages	Part / Chapter	Topic / What it covers
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p26–27	PART V	INSTRUMENTS AND WHAT THEY MEASURE — <i>Instruments & What They Measure</i> — <i>barometer, seismograph, etc.</i>
p28–29	PART W	DISCOVERIES AND INVENTIONS IN PHYSICS — <i>Discoveries & Inventions</i> — <i>Newton, Faraday, Chadwick.</i>
p30–31	PART X	DAILY-LIFE PHYSICS ("WHY" QUESTIONS) — <i>Daily-Life Physics</i> — <i>pressure cooker, blue sky, ice floats.</i>
p32–35	PART AA	KEY CONSTANTS AND FORMULAS — <i>Key Constants & Formulas</i> — <i>single-page exam-day reference.</i>

PART A — MECHANICS

Chapter A1 — SI Units

The scientific world agreed on one system of measurement — the **International System of Units (SI)**. Here are its 7 base units, which all other units are built from.

Quantity	Unit	Symbol
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric current	ampere	A
Temperature	kelvin	K
Amount of substance	mole	mol
Luminous intensity	candela	cd

REMEMBER

"My Kind Sister Always Keeps Many Candles" — Mass (kg), Kind (K = kelvin), Sister (s = second), Always (A = ampere), Keeps (this is a connector), Many (m = metre), Candles (cd = candela). Or more directly: **metre, kilogram, second, ampere, kelvin, mole, candela** — "Mr. K Sees Amphibians Killing Many Critters."

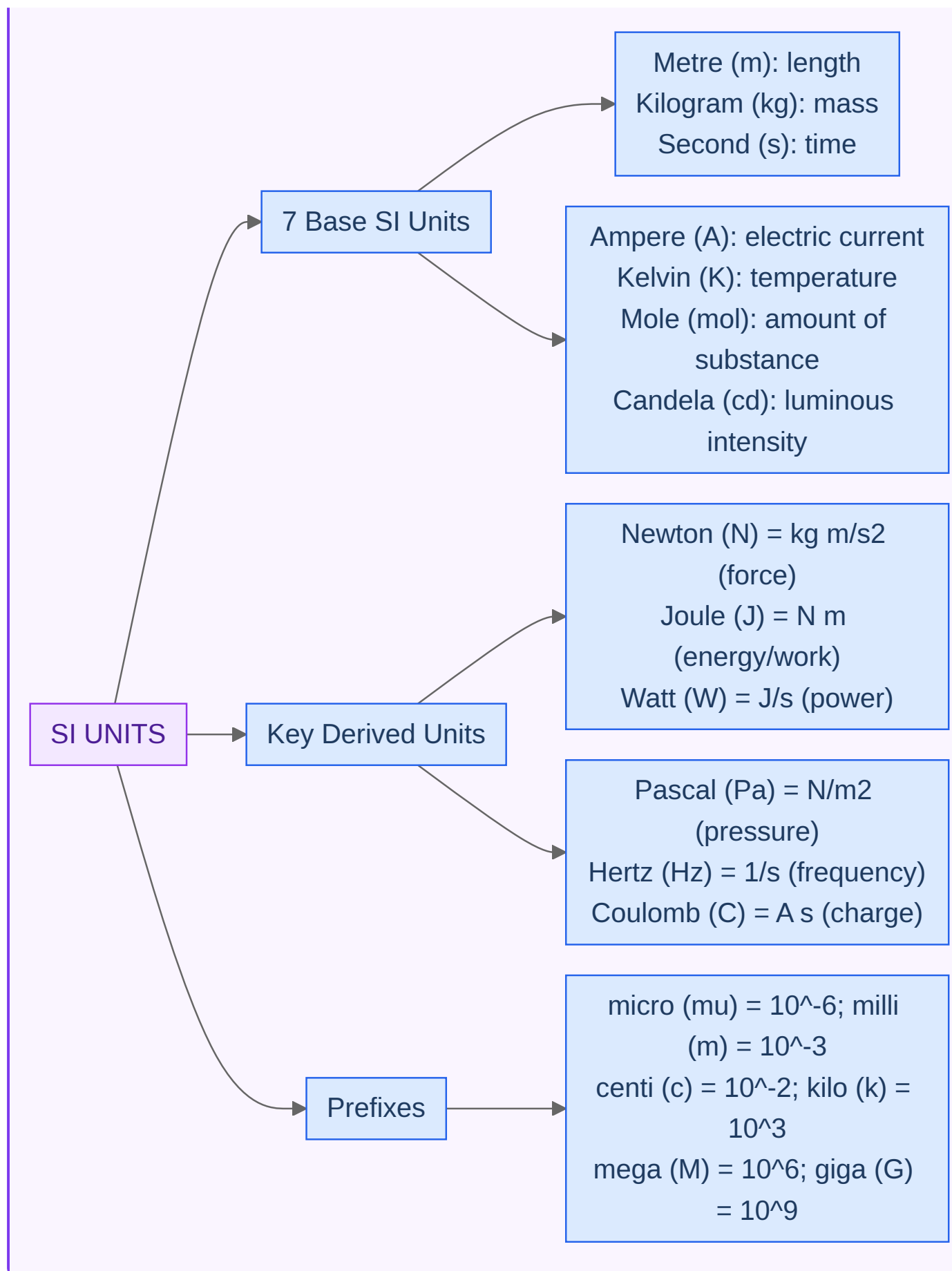
Derived Units to Know

Quantity	Unit	Symbol	In base units
Force	newton	N	$\text{kg}\cdot\text{m}/\text{s}^2$
Pressure	pascal	Pa	N/m^2
Energy / Work	joule	J	$\text{N}\cdot\text{m}$
Power	watt	W	J/s
Frequency	hertz	Hz	s^{-1}
Electric charge	coulomb	C	$\text{A}\cdot\text{s}$
Voltage	volt	V	J/C
Resistance	ohm	Ω	V/A

KEY FACT

The commercial unit of electrical energy is 1 kilowatt-hour (kWh) = 1 unit on your electricity bill = 3.6×10^6 joules. 1 horsepower = 746 watts (not 750 — examiners use this to catch those who round up carelessly).

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter A2 — Newton's Laws of Motion
Isaac Newton, 1687 — published in *Principia Mathematica*.

- Governed everything from cricket balls to rockets.
- Refined by Einstein for extreme speeds (relativity) and quantum scales.

KEY FACT

Quick Recall — 3 Laws in 3 words: Inertia | $F=ma$ | Reaction. Every exam scenario maps to exactly one of these. "Lurching / jerking" = 1st Law. "Greater force = greater push" = 2nd Law. "Rocket / gun / swimming" = 3rd Law.

First Law — Law of Inertia

Statement:

- An object **at rest stays at rest**.
- An object **moving at constant velocity** continues at the same velocity.
- Unless acted upon by an **external force**.

Inertia:

- Tendency to resist change in motion.
- **Proportional to mass** — a truck is harder to start moving than a bicycle.

EXAM ALERT

Exam applications of Newton's First Law: seat belts (body wants to continue moving when car stops), pulling a tablecloth quickly (plates stay due to inertia), coins on a card flicked away (coins fall into the glass), dust flying off a beaten carpet. Any "lurching" or "jerking" scenario = First Law.

Second Law — $F = ma$

Statement: rate of change of momentum $\propto$ applied force, in the direction of the force.

Simplified: Force = mass $\times$ acceleration

- More force $\rightarrow$ more acceleration.
- More mass $\rightarrow$ less acceleration for the same force.
- SI unit: **newton (N)**. 1 N accelerates 1 kg at 1 m/s^2 .

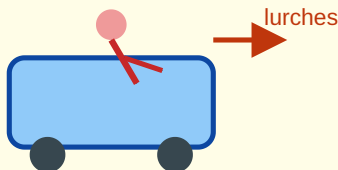
Third Law — Action and Reaction

Statement: for every action there is an **equal and opposite reaction**.

- The two forces act on **different objects** (this trips up most students).

Newton's 3 Laws — Visual Summary

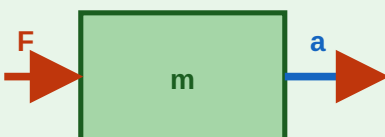
1st: Inertia



Car brakes suddenly →
body continues forward
(why seatbelts!)

*Object resists change
in its state of motion.*

2nd: $F = ma$

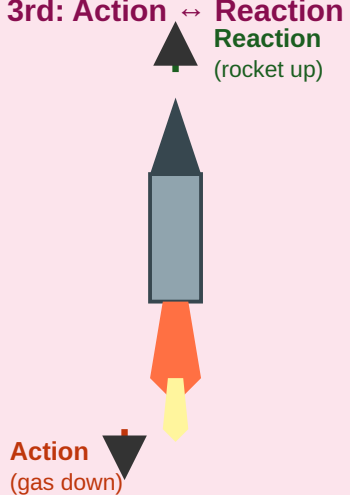


$F = m \times a$

More $F \rightarrow$ more a
More $m \rightarrow$ less a (same F)

$1\text{ N} = \text{mass} \times \text{accel}$
(of $1\text{ kg} @ 1\text{ m/s}^2$)

3rd: Action ↔ Reaction



Action
(gas down)

Reaction
(rocket up)

EXAM ALERT

Third Law applications that appear in exams: **rocket propulsion** (gas pushed backward → rocket pushed forward), **gun recoil** (bullet pushed forward → gun pushes back), **swimming** (you push water backward → water pushes you forward), **helicopter tail rotor** (counters torque from main rotor). Jet engines, walking, and even rowing a boat are Third Law applications.

KEY FACT

Analogy: The Third Law is like clapping — your left hand hits the right hand with exactly the same force the right hand hits the left, but both hands feel it. The forces are equal and opposite but act on different objects — just like rocket gas and rocket body.

EXAM ALERT

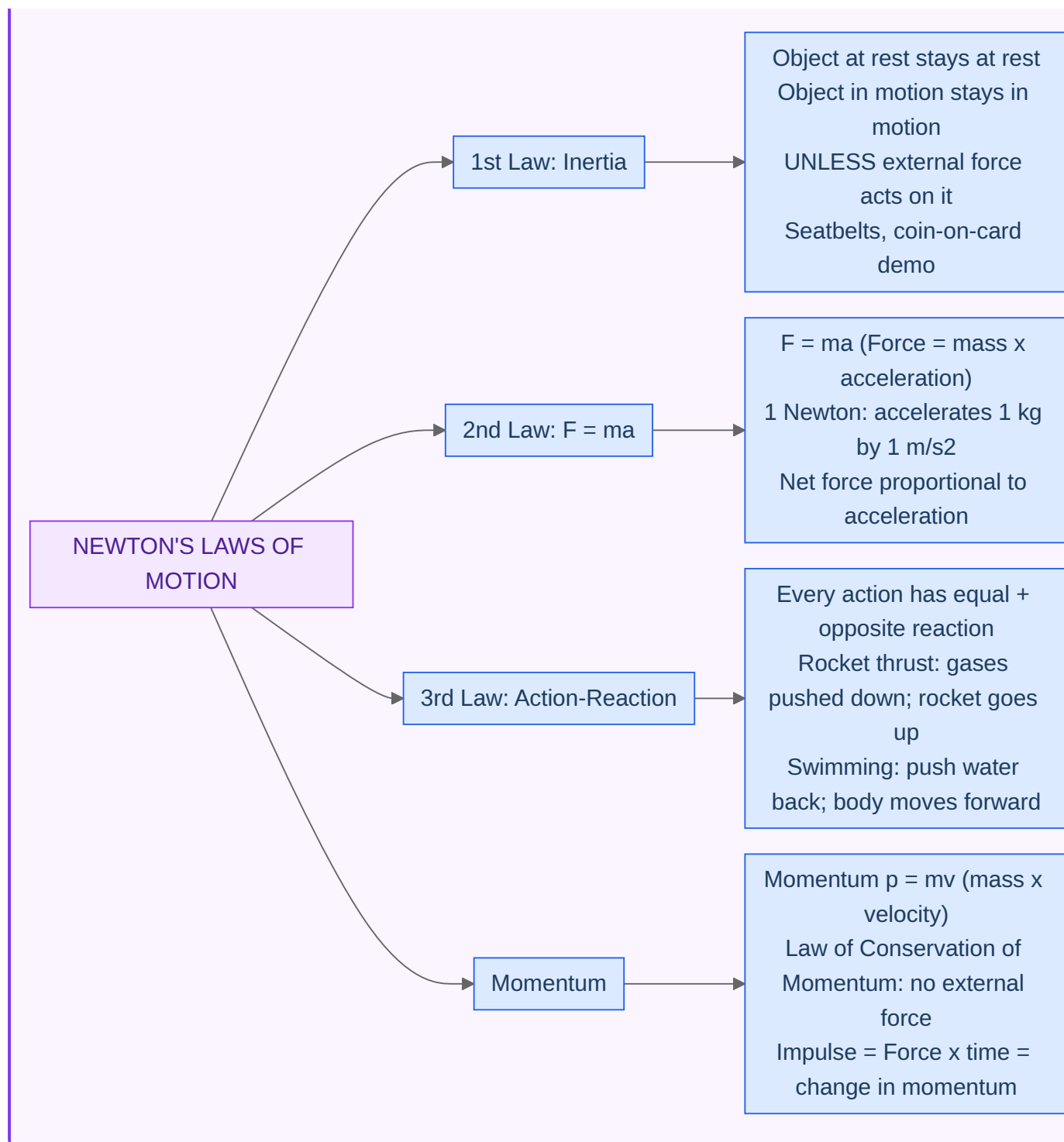
⚠ TRAP: Newton's Third Law says the reaction force is equal and opposite — but students often think the two forces cancel. They CANNOT cancel because they act on **different objects**. A book on a table: book pushes table down (action), table pushes book up (reaction). These don't cancel — they act on different bodies.

Equations of Motion (Constant Acceleration)

- $v = u + at$
- $s = ut + \frac{1}{2}at^2$
- $v^2 = u^2 + 2as$

Where: u = initial velocity, v = final velocity, a = acceleration, t = time, s = displacement.

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter A3 — Gravitation

Newton's Universal Law

Every two masses attract each other with a force proportional to the product of their masses and inversely proportional to the square of the distance between them.

$$F = G \times (m_1 \times m_2) / r^2$$

$G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ (Cavendish first measured this in 1798).

Acceleration Due to Gravity (g)

- g at Earth's surface = 9.8 m/s² (use 10 for quick calculations).

- **g is less at the equator** than at the poles — because: (1) the Earth is oblate (bulges at equator, so equatorial surface is farther from the centre), and (2) rotational effect reduces effective gravity.
- **g decreases** as you go higher (away from centre).
- **g decreases** as you go deeper (less mass pulling you).
- **At Earth's centre, $g = 0$.**

KEY FACT

Escape velocity from Earth = **11.2 km/s** (~40,300 km/h). This is the minimum speed an object needs to permanently escape Earth's gravity without further propulsion. Orbital velocity at Earth's surface ≈ 7.9 km/s — the speed needed to maintain a circular orbit just above the surface.

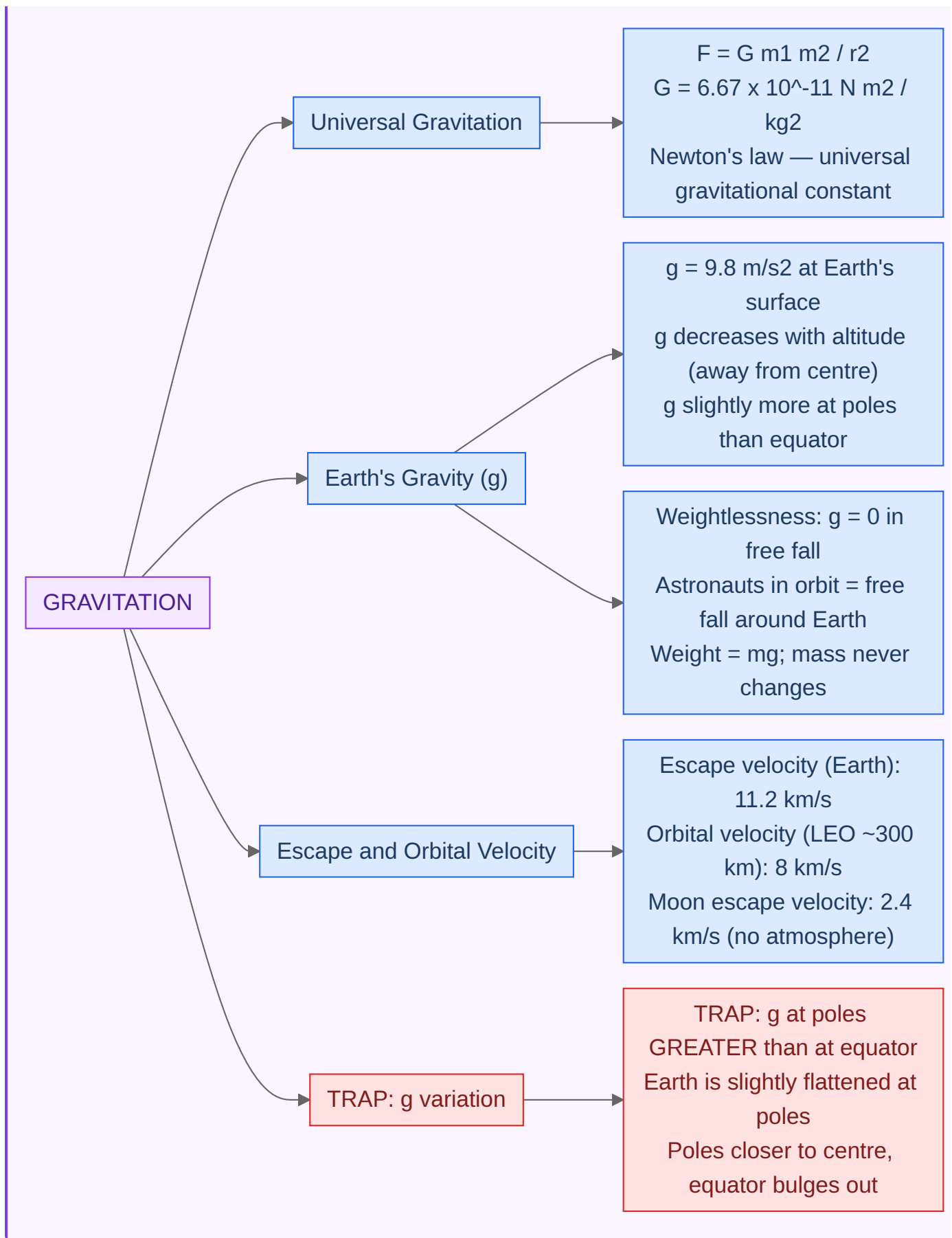
Kepler's Laws of Planetary Motion

1. **Law of Orbits** — Planets move in elliptical orbits with the Sun at one focus.
2. **Law of Areas** — A line joining a planet to the Sun sweeps out equal areas in equal time intervals (planets move faster when closer to the Sun).
3. **Law of Periods** — $T^2 \propto r^3$ (square of orbital period is proportional to cube of semi-major axis).

Satellites

- A **geostationary satellite** orbits at ~36,000 km above the equator, completing one orbit in exactly 24 hours — so it appears stationary relative to Earth's surface. Used for telecommunications and weather.
- **Weightlessness** in a satellite — the satellite and everything inside are in free fall together. The apparent weight is zero because both the person and the floor accelerate downward at the same rate.

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter A4 — Work, Energy, Power

- Work** = Force $\times$ displacement $\times \cos(\theta)$. Work is done only when displacement occurs in the direction of force. Unit: joule (J).

- **Kinetic Energy (KE)** = $\frac{1}{2}mv^2$
- **Potential Energy (gravitational PE)** = mgh
- **Power** = Work done / Time taken. Unit: watt (W). $P = F \times v$.

Conservation of Energy

The law:

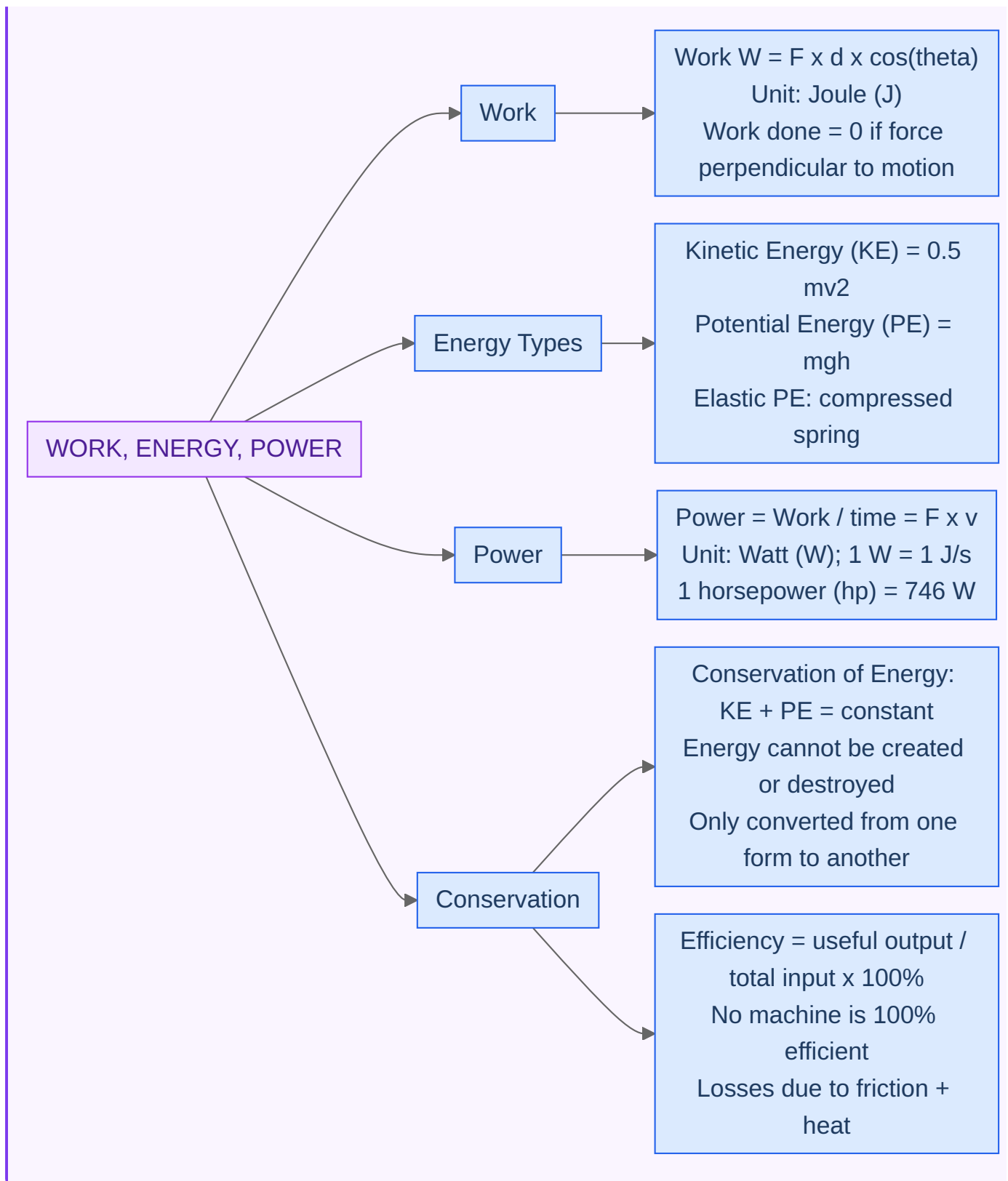
- Energy **cannot be created or destroyed**.
- It only **transforms** from one form to another.
- Total energy of an **isolated system is constant**.

Classic example — the pendulum:

- At the **highest point** → maximum PE, zero KE.
- At the **lowest point** → zero PE, maximum KE.
- (PE + KE) stays constant throughout the swing (ignoring air resistance).

PART B — PROPERTIES OF MATTER

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Chapter B1 — Pressure, Fluids, Buoyancy

- **Pressure** = Force / Area. SI unit: pascal (Pa). A sharp knife cuts better than a blunt one because the same force is applied over a smaller area.
- **Atmospheric pressure** at sea level = 101,325 Pa $\approx$ 1 atm = 760 mm Hg.
- Atmospheric pressure decreases with altitude (less air column above).

Pascal's Law

Statement: pressure on an enclosed fluid transmits **equally in all directions**.

Why it matters:

- Hydraulic systems can **multiply force** enormously from a small input.
- Applications — car brakes, JCB excavators, dental chairs, hydraulic press.

Archimedes' Principle

Statement: upward buoyant force on a submerged body = **weight of fluid displaced**.

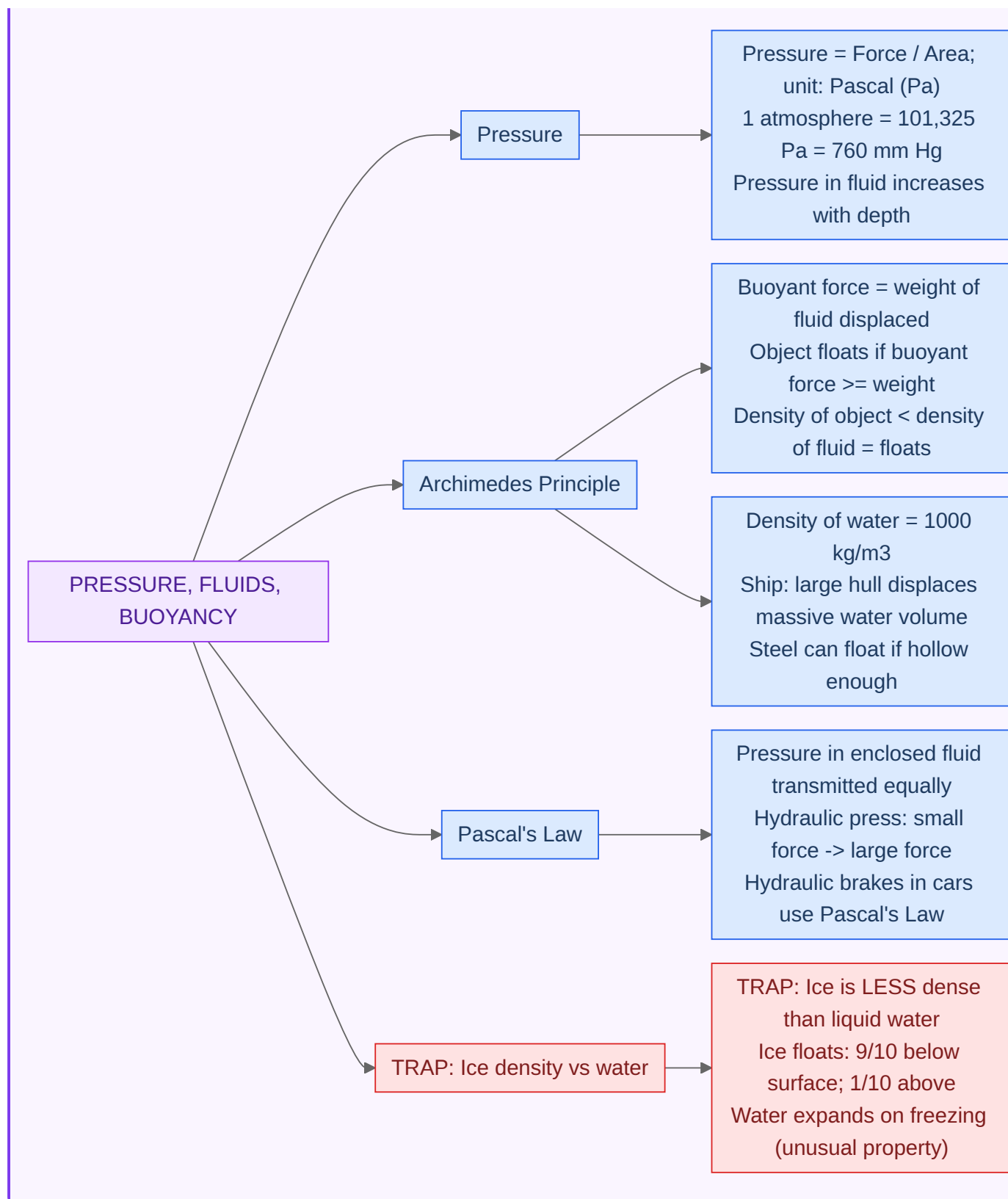
Steel ship vs steel ball:

- A solid steel ball **sinks** — density of steel $\gg$ density of water.
- A steel ship **floats** because it is **hollow**.
- Effective density (steel + interior air) $<$ density of water.
- Ship displaces enough water for the buoyant force to equal its weight.

EXAM ALERT

Ice floats on water — this is anomalous expansion. Water expands on freezing (unusual behaviour). Ice density is 0.92 g/cm^3 vs water at 1.0 g/cm^3 . This is why fish survive winter — ice floats and forms an insulating layer; liquid water at 4°C stays below, where fish live.

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter B2 — Surface Tension, Viscosity, Capillarity

Surface tension

- Why water forms droplets — spherical shape = minimum surface area for a given volume.
- Why insects can walk on water; why a needle can float on still water.
- **Detergent reduces surface tension** — soap molecules disrupt the surface layer.

Viscosity

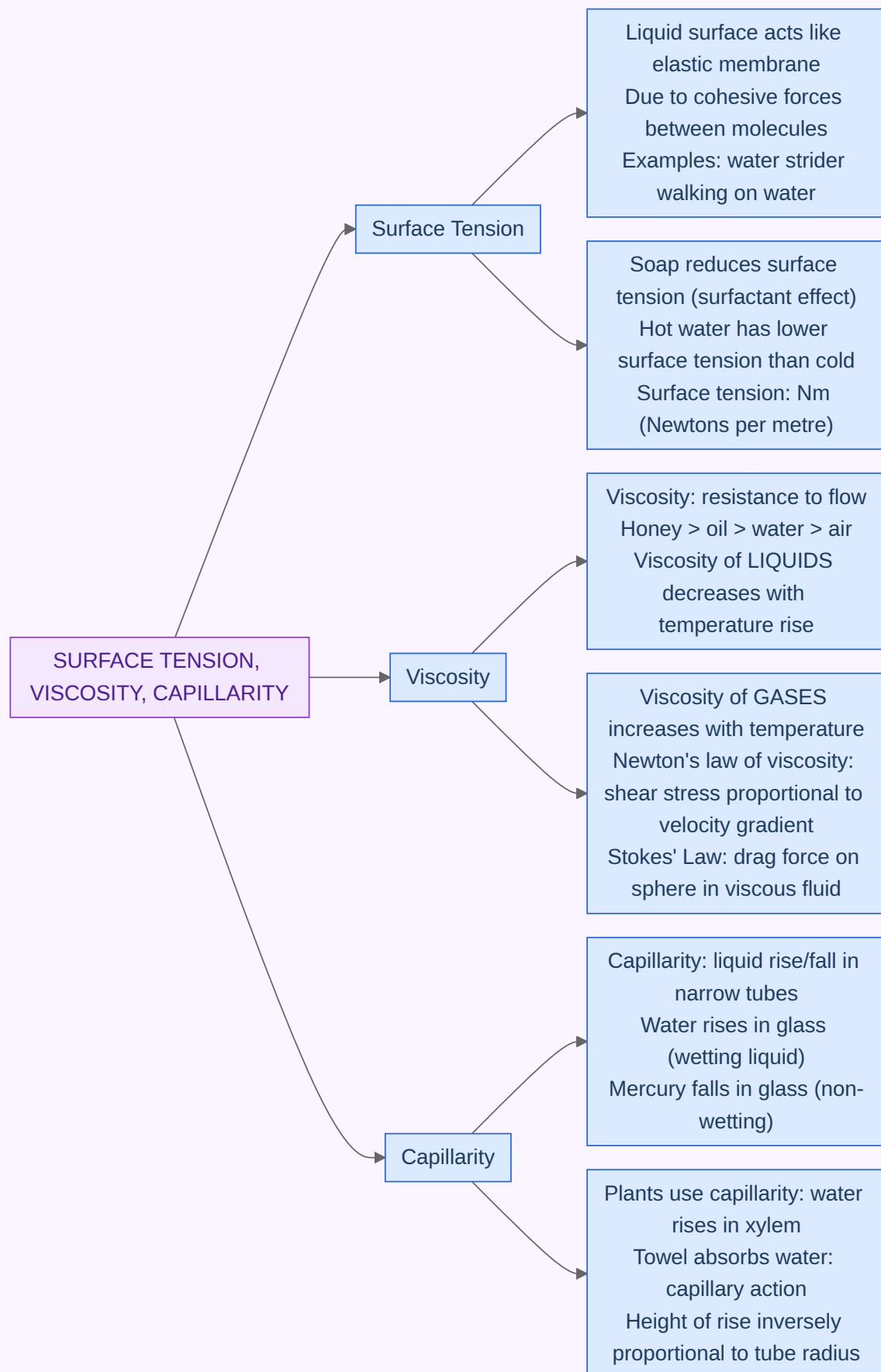
- Internal friction in fluids; resistance to flow.
- Ranking: **honey > water > petrol**.
- **Viscosity decreases with temperature** (why engine oil thins when hot).

Capillarity

- Rise or fall of liquid in narrow tubes — surface tension vs gravity.
- Water **rises** in a glass capillary (adhesion to glass > cohesion).
- Mercury **falls** in a glass capillary (cohesion > adhesion).
- Plants use capillary action to draw water up roots and stems.

PART C — HEAT AND THERMODYNAMICS

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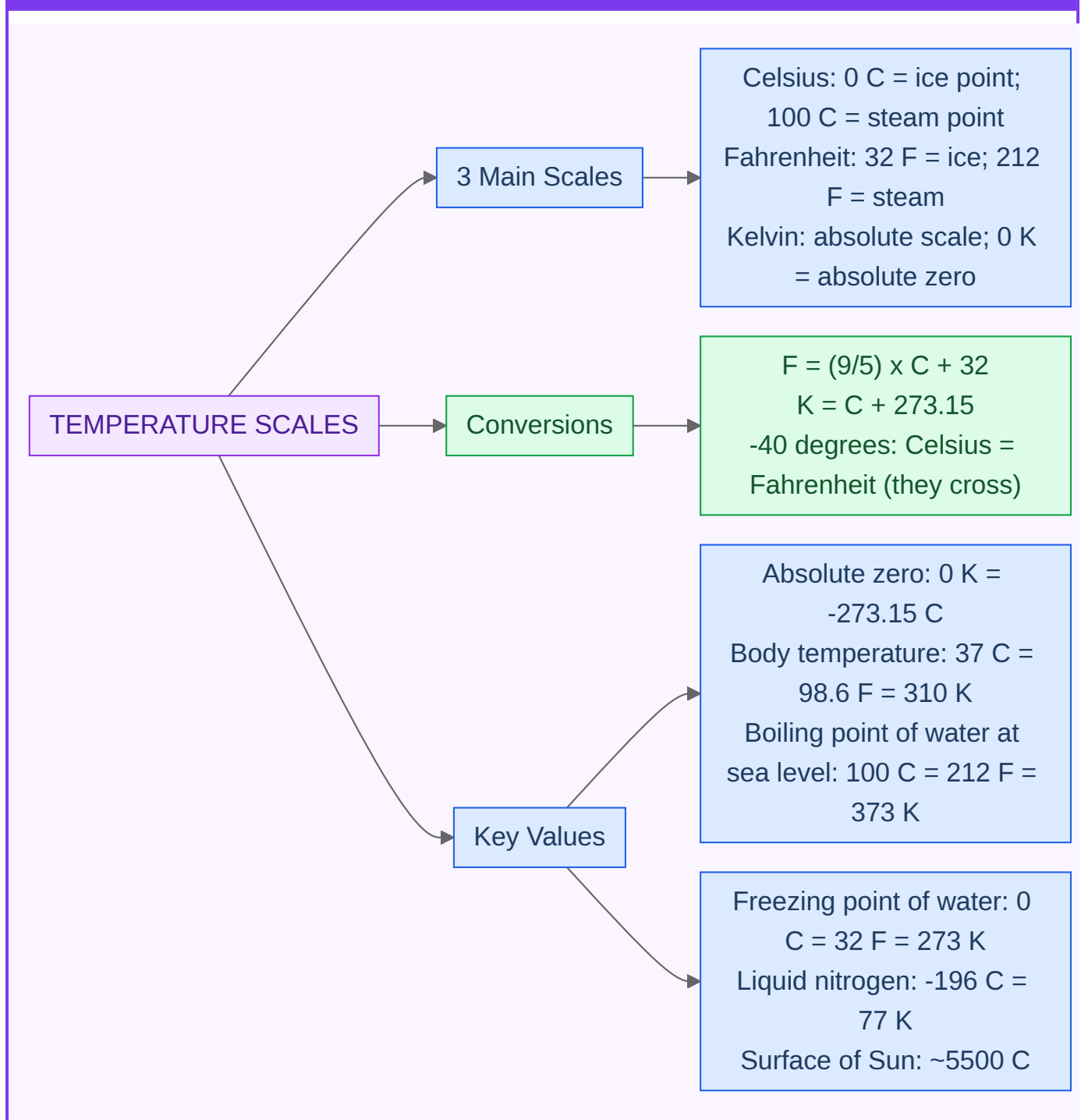
Chapter C1 — Temperature Scales

Scale	Freezing of water	Boiling of water
Celsius (°C)	0	100
Fahrenheit (°F)	32	212
Kelvin (K)	273.15	373.15

Conversion: $^{\circ}\text{F} = (9/5 \times ^{\circ}\text{C}) + 32$. Kelvin = Celsius + 273.15.

Absolute zero ($0\text{ K} = -273.15\text{ }^{\circ}\text{C}$) — theoretically the coldest possible temperature; particles have minimum possible energy.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



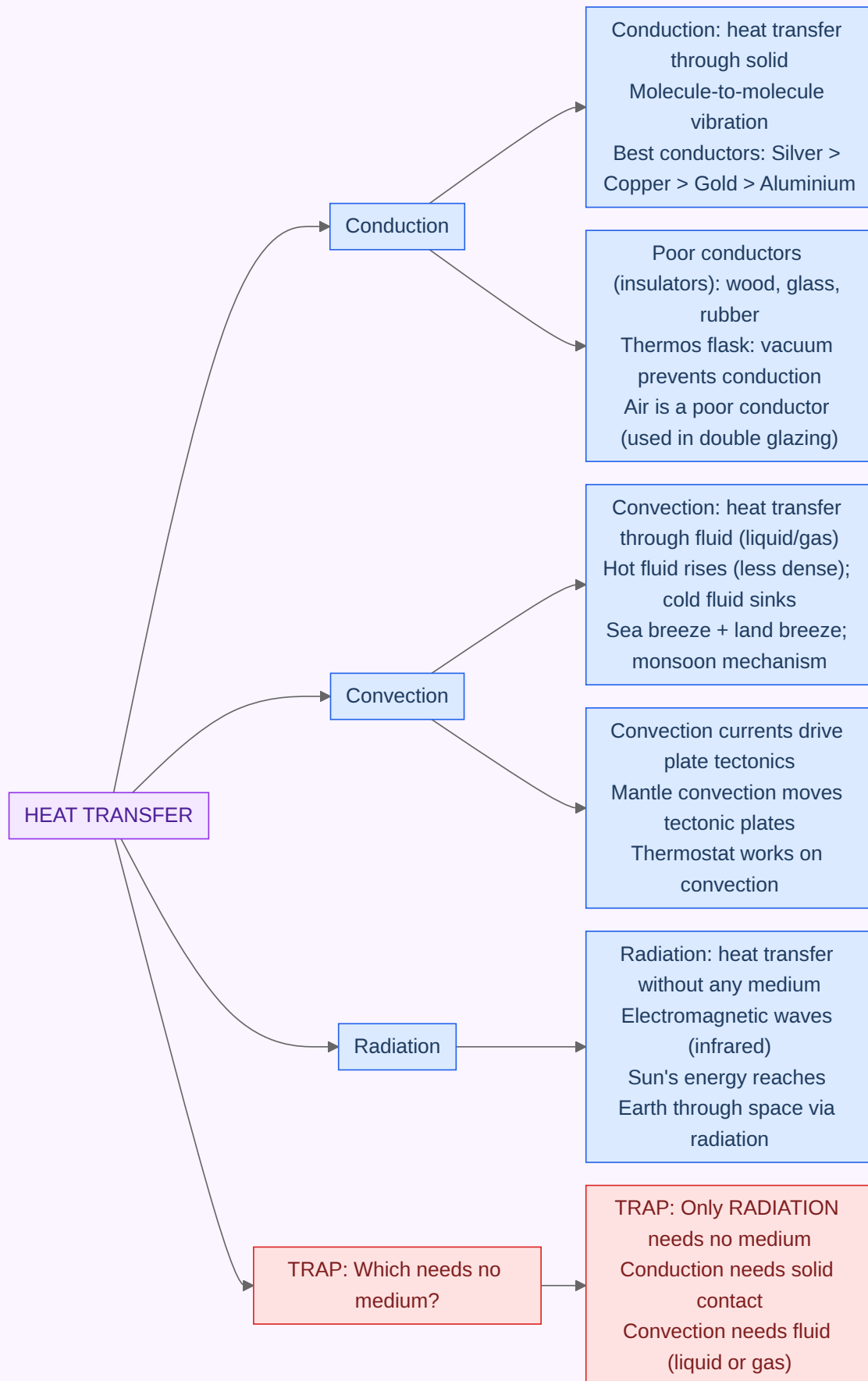
Chapter C2 — Heat Transfer

Mode	Medium needed	Mechanism	Examples
Conduction	Solid (mainly)	Particle vibration, electron flow	Metal rod heating, iron; ice cube melting in hand
Convection	Fluid (liquid/gas)	Bulk movement of fluid	Boiling water, sea breeze, room air conditioning
Radiation	None — vacuum OK	Electromagnetic waves	Sun warming Earth, warmth from a fire

KEY FACT

The Sun's heat reaches Earth through the vacuum of space — this is radiation (electromagnetic waves). Only radiation requires no medium. This is why a thermos (vacuum flask) has a silvered interior — it prevents heat loss by all three mechanisms: vacuum stops conduction and convection, silvering reflects radiation.

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter C3 — Specific Heat and Latent Heat

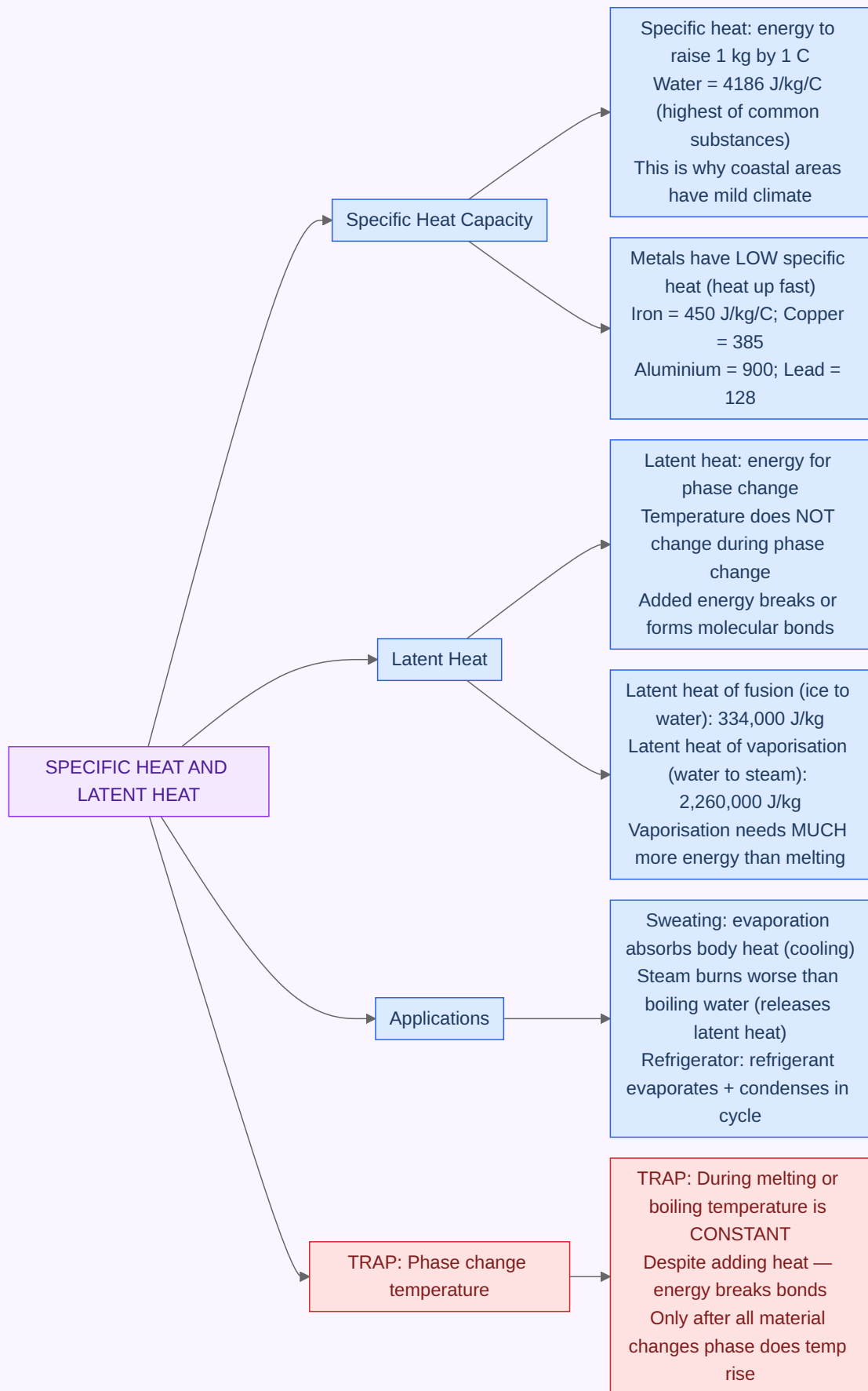
- **Specific heat** — heat energy needed to raise 1 kg of a substance by 1 °C (or 1 K).

- Water: $4184 \text{ J/kg}\cdot^\circ\text{C}$ — the highest among common substances. This is why coastal climates are milder — oceans absorb and release enormous heat with little temperature change.
- Iron: $\sim 449 \text{ J/kg}\cdot^\circ\text{C}$. Aluminium: $\sim 900 \text{ J/kg}\cdot^\circ\text{C}$.
- **Latent heat** — heat absorbed or released during a phase change (state change) WITHOUT any temperature change.
- Latent heat of fusion of ice: 334 J/g (ice $\rightarrow$ water at 0°C).
- Latent heat of vaporisation of water: 2260 J/g (water $\rightarrow$ steam at 100°C).

EXAM ALERT

Steam burns are worse than boiling water burns at the same 100°C — because steam releases its latent heat of vaporisation (2260 J/g) when it condenses on your skin, in addition to the heat from cooling. We sweat to cool down because sweat evaporating absorbs latent heat of vaporisation from the skin surface.

🕒 CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



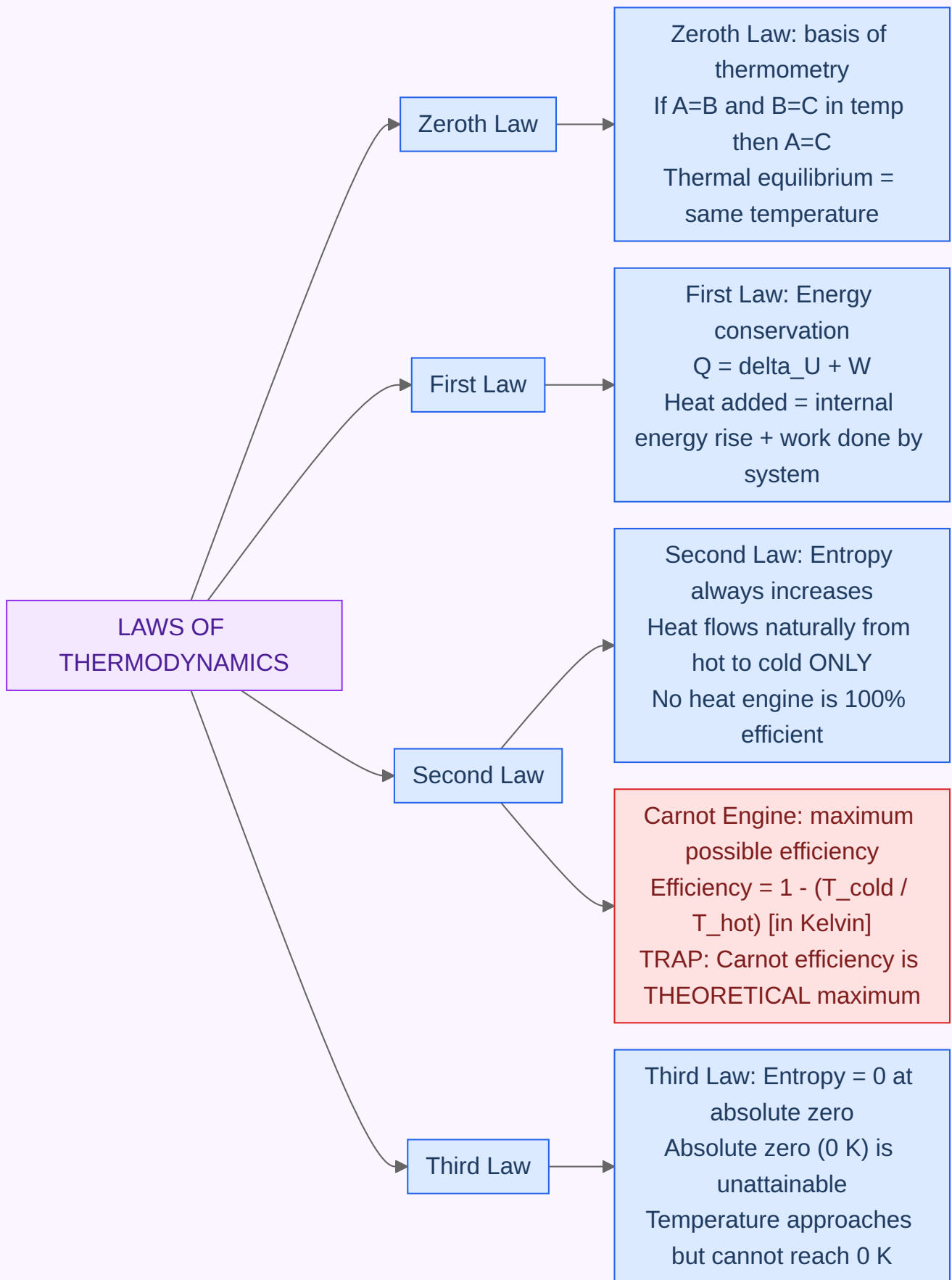
Chapter C4 — Laws of Thermodynamics

1. **Zeroth Law** — If system A is in thermal equilibrium with system B, and B with C, then A and C are also in equilibrium. This defines temperature.
2. **First Law** — $\Delta U = Q - W$. Energy is conserved. Heat added to a system equals change in internal energy plus work done by the system.
3. **Second Law** — Heat cannot spontaneously flow from cold to hot. Entropy (disorder) of an isolated system always increases.
4. **Third Law** — At absolute zero, entropy of a perfect crystal is zero. Absolute zero is unreachable in practice.

A refrigerator moves heat from cold to hot — but only by consuming electricity (external work). This is the Second Law in action.

PART D — WAVES, SOUND, AND LIGHT

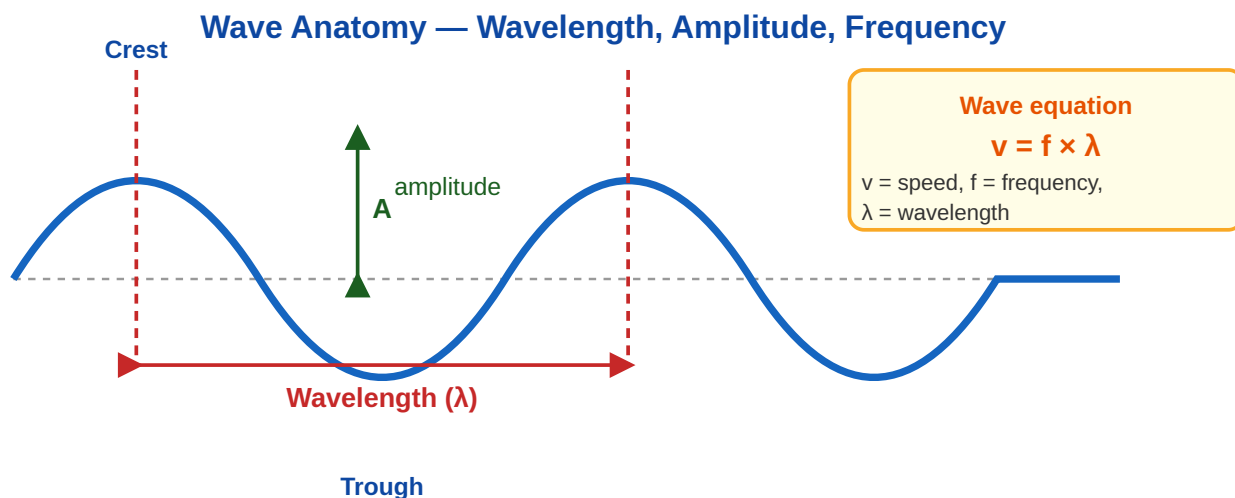
⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D1 — Sound

Sound is a mechanical wave:

- It needs a **medium** to travel (solid / liquid / gas).
- It **cannot travel through vacuum** — that's why space is silent.
- It is a **longitudinal** wave — compressions + rarefactions along the direction of travel.



Speed of Sound

Medium	Speed (approximate)
Air at 0 °C	331 m/s
Air at 20 °C	343 m/s
Water	~1500 m/s
Steel	~5000 m/s

Order of speeds: Solid > Liquid > Gas (because molecules are closer; vibrations transmit faster).

Speed increases with temperature (in air, ~0.6 m/s per °C rise).

REMEMBER

"Soldiers Lightly Go" — Solid → Liquid → Gas: speed of sound decreases in this order. Solid is fastest because particles are packed tightly and vibrations transfer instantly.

EXAM ALERT

⚠ TRAP: Sound travels **FASTEST** in solids, **SLOWEST** in gases — opposite of what many students assume. "Sound must travel faster in open air (gas) than in dense iron (solid)" is wrong. Iron: ~5000 m/s; Air: 343 m/s. Denser and more elastic medium = faster sound.

Frequency Ranges

Range	Frequency	Who uses it
Infrasonic	Below 20 Hz	Whales, elephants, earthquakes
Audible (human)	20 Hz to 20,000 Hz	Humans
Ultrasonic	Above 20,000 Hz	Bats, dolphins, dogs; medical sonography, SONAR

KEY FACT

Echo needs a minimum distance of **17 metres** between you and the reflecting surface. Why? Sound takes 0.1 seconds to reach the reflector and return (minimum time for the human ear to distinguish echo from original). At 343 m/s, that is $343 \times 0.1 / 2 = 17.15$ m.

Doppler Effect

The idea: apparent frequency changes when source and observer are in relative motion.

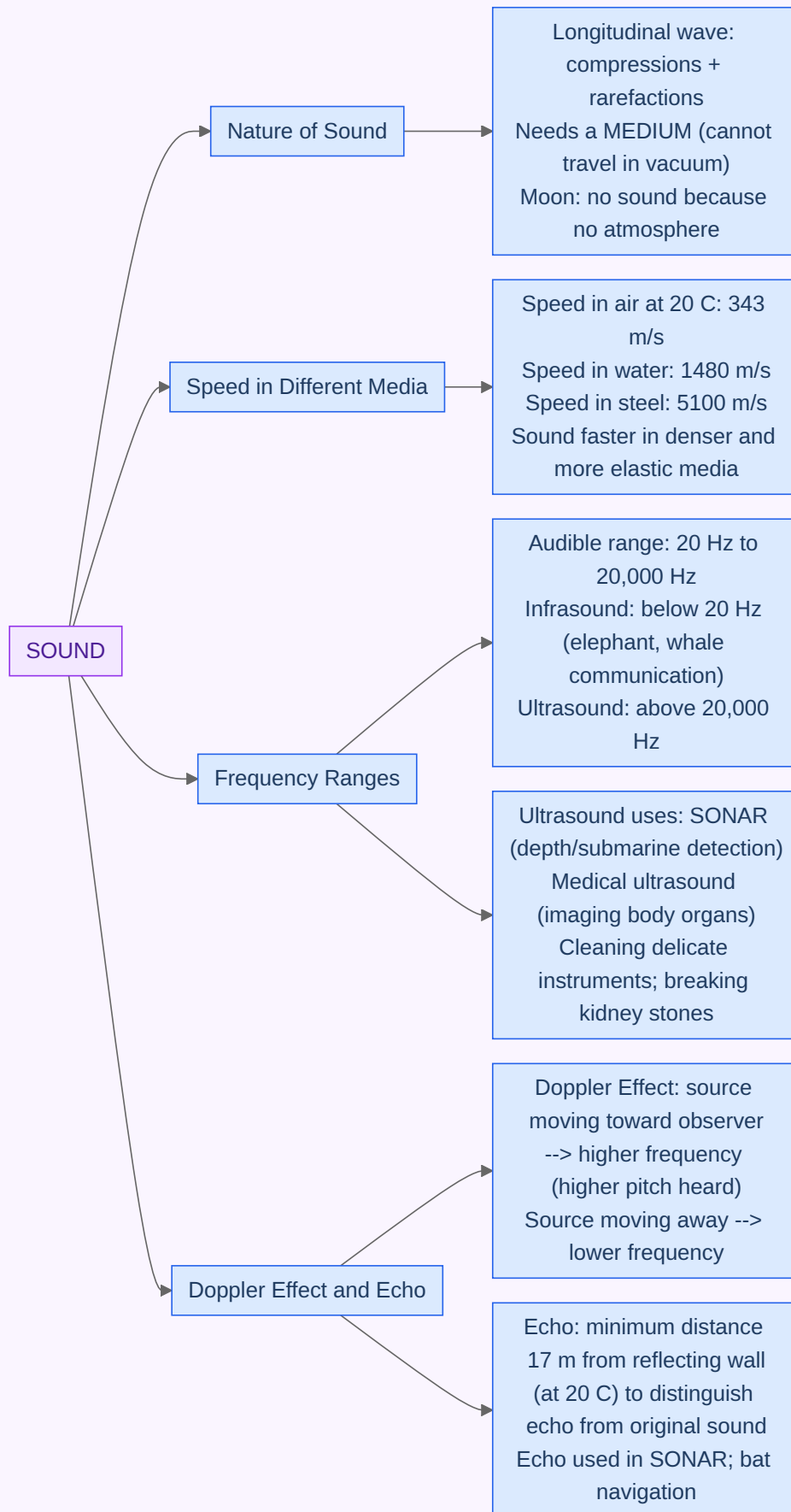
- **Source approaches** → frequency appears **higher** (pitch rises).
- **Source moves away** → frequency appears **lower** (pitch falls).
- **Classic example:** an ambulance siren rising in pitch as it approaches, dropping as it passes.

SONAR

Sound Navigation And Ranging.

- Uses **ultrasonic pulses** to detect underwater objects.
- Maps the ocean floor; finds submarines, shoals of fish.
- **Distance = (speed of sound in water × echo time) / 2** (divide by 2 — wave goes and returns).

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter D2 — Light and Optics

Nature of light:

- Light is an **electromagnetic wave**.
- Speed in vacuum: **$c = 3 \times 10^8 \text{ m/s}$** — the fastest speed in the universe.
- **Slows down in denser media** — water $\sim 2.25 \times 10^8 \text{ m/s}$, glass $\sim 2 \times 10^8 \text{ m/s}$.

Refractive index (n):

- **$n = \text{speed in vacuum} / \text{speed in medium}$** .
- Higher $n \rightarrow$ light slows more $\rightarrow$ bends more.
- **Glass ≈ 1.5 , water ≈ 1.33 , diamond ≈ 2.42 .**

Reflection

- Angle of incidence = angle of reflection.
- Incident ray, reflected ray, and normal are in the same plane.

Refraction (Snell's Law)

Snell's law: $n_1 \sin \theta_1 = n_2 \sin \theta_2$

- Light **bends toward the normal** when entering a **denser** medium.
- Light **bends away from the normal** when entering a **less-dense** medium.

Total Internal Reflection (TIR)

- Happens when light goes **denser $\rightarrow$ less-dense** at angle **$>$ critical angle**.
- Light is **completely reflected back** into the denser medium.
- **Applications:**
 - **Diamond sparkle** — critical angle $\sim 24^\circ \rightarrow$ most rays bounce inside.
 - **Optical fibres** — internet backbone; light bounces along the core.
 - **Mirages** — hot air over road acts as less-dense layer.

Mirrors — Which Type Does What

Mirror	Image formed	Used in
Plane	Same size, virtual, erect, laterally inverted	Looking glass
Concave (converging)	Depends on object position; real+inverted if beyond focus	Shaving mirror, headlights, dentist's mirror, reflecting telescopes
Convex (diverging)	Always virtual, erect, diminished	Rear-view mirrors in vehicles, security mirrors

EXAM ALERT

Rear-view mirrors in vehicles are always convex — they give a wider field of view (the image is diminished but you see more). The "objects in mirror are closer than they appear" warning is because the convex mirror makes distant objects appear farther away. A driver checking a concave rear-view mirror would get a misleading sense of distances.

Lenses — Which Type Does What

Lens	Image type	Used for
Convex (converging)	Real or virtual depending on position	Magnifying glass, cameras, projectors, microscopes, eye's own lens
Concave (diverging)	Always virtual, erect, diminished	Correcting myopia (short-sightedness)

Lens formula: $1/v - 1/u = 1/f$. Power of lens: $P = 1/f$ (metres). Unit: dioptre (D).

Defects of Vision and Corrections

Defect	What happens	Correction
Myopia (short-sightedness)	Image forms in front of retina; can see near but not far	Concave (diverging) lens
Hypermetropia (long-sightedness)	Image forms behind retina; can see far but not near	Convex (converging) lens
Presbyopia	Eye lens stiffens with age; cannot adjust	Bifocal lens
Astigmatism	Cornea unevenly curved	Cylindrical lens
Cataract	Lens becomes opaque	Surgery to replace lens

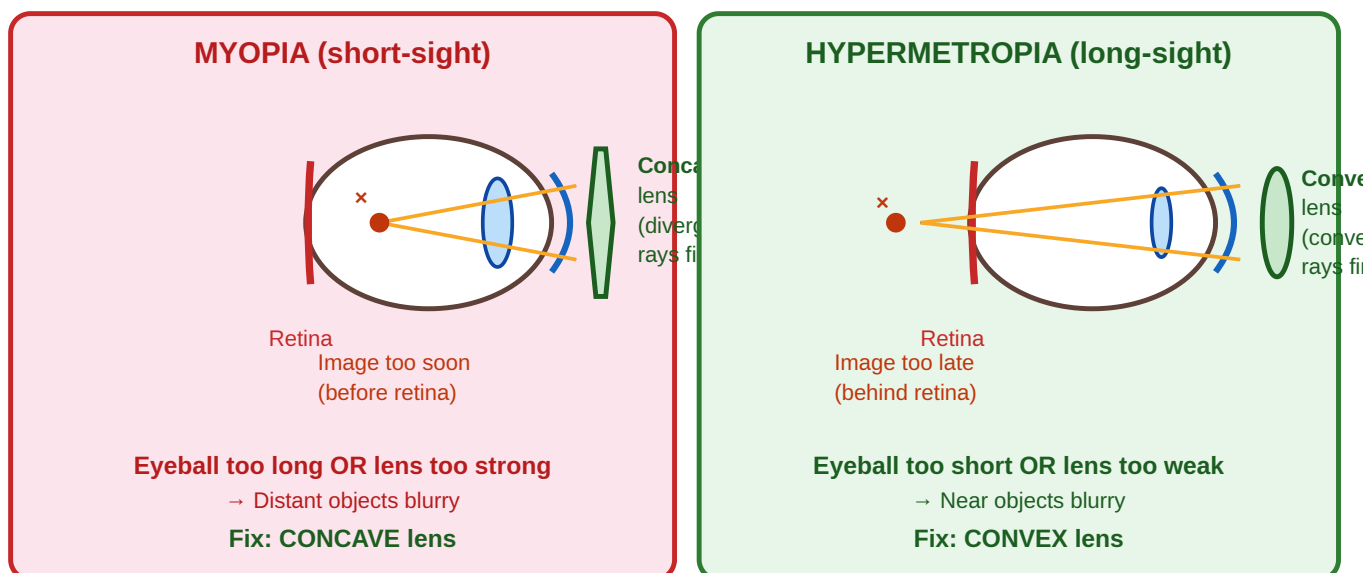
KEY FACT

Exam shortcut — Vision defects in 4 words: Myopia = **Concave** | Hypermetropia = **Convex**. Quick memory: **My Cave** (Myopia → Concave) and **Hyper Vex** (Hypermetropia → Convex). Concave diverges light so the image moves back onto the retina. Convex converges light so the image moves forward onto the retina.

EXAM ALERT

⚠ TRAP: "Concave lens is used to correct which defect?" — Answer is **Myopia** (near-sighted, cannot see far). Many students mix this up and write Hypermetropia. Use: **Concave = Cannot see far = Myopia**. Convex = Hypermetropia (cannot see near). Examiners set the question both ways.

Vision Defects + Lens Corrections



Memory aid: M → C (Myopia → Concave) | Hyper → Cv (Hypermetropia → Convex).

REMEMBER

Myopia → Concave, Hypermetropia → Convex. Memory trick: the words **Myopia** and **Concave** both have fewer common letters to remember — just link the **Myopic** person needs a **Concave** lens. Or: Myopia = "My can't see far" → need to diverge the light more → concave.

Dispersion and Colour

Dispersion

- White light splits into its spectrum **VIBGYOR** when passing through a prism.
- Newton first demonstrated this in **1666**.

VIBGYOR — order matters

- **Violet** → Indigo → **Blue** → **Green** → **Yellow** → **Orange** → **Red**.
- **Decreasing wavelength** (and **increasing frequency**) going left to right.
- **Violet** has the highest frequency in the visible range.

Rainbow

- Caused by **refraction + total internal reflection + dispersion** inside raindrops.
- Always opposite the sun; primary bow is 42° from the anti-solar point.

Blue sky — Rayleigh scattering

- Shorter (blue) wavelengths scatter **~5× more** than longer (red) wavelengths.
- Sunlight enters the atmosphere → blue scatters in all directions → whole sky looks blue.

Red sunset

- At sunrise / sunset, sunlight travels through **much more atmosphere**.
- Blue has already scattered away en route.
- **Red**, which scatters least, dominates → red-orange sky.

KEY FACT

Diamond sparkles because its refractive index is very high (2.42), giving a very small critical angle (~24°). Most light rays entering the diamond hit a face at more than 24° and undergo total internal reflection repeatedly — they bounce inside the diamond and finally exit from the top. This internal bouncing creates the famous sparkle.

Electromagnetic Spectrum

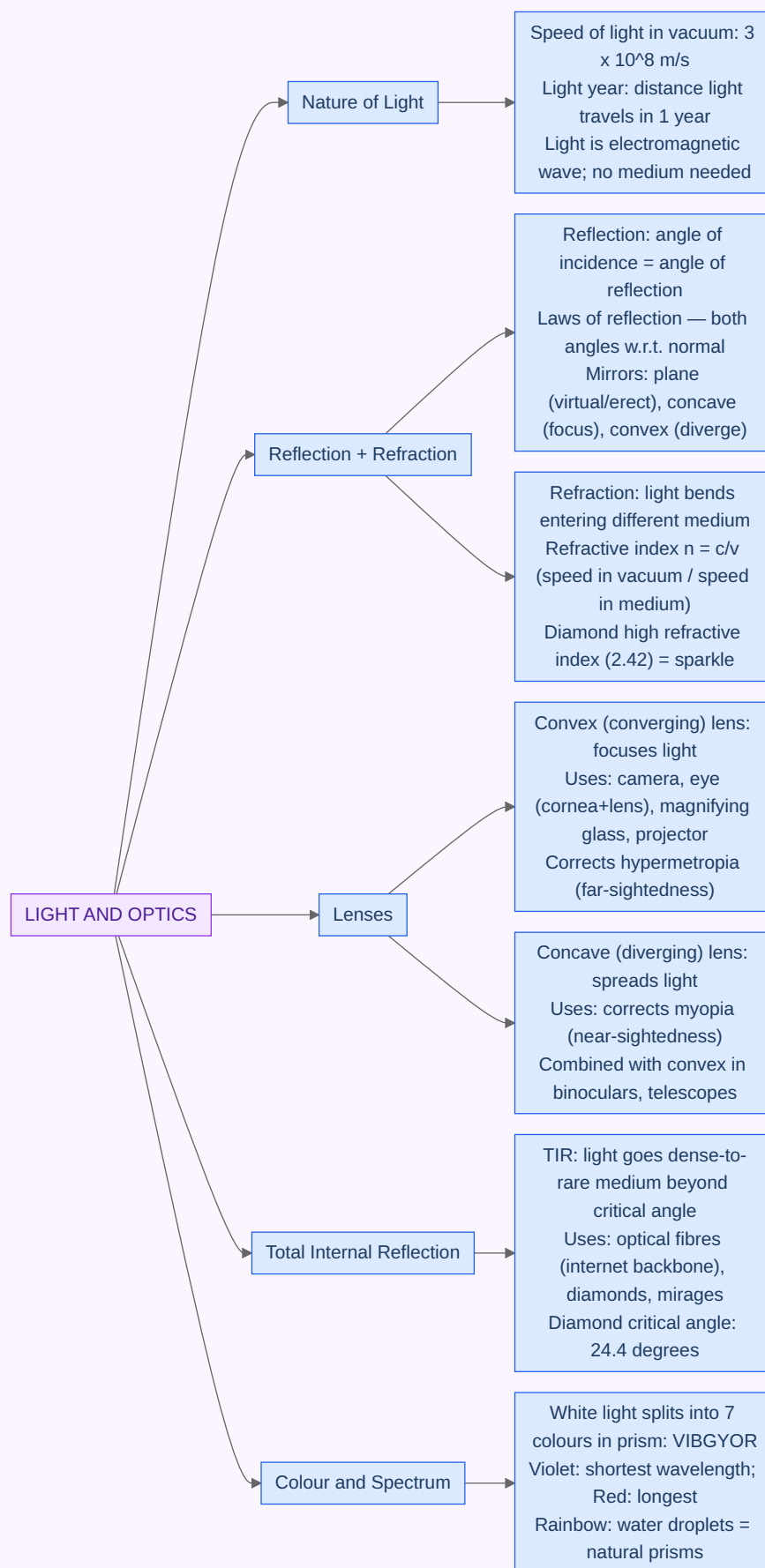
From lowest to highest frequency (highest to lowest wavelength):

Radio → Microwaves → Infrared → Visible light → Ultraviolet → X-rays → Gamma rays

Band	Wavelength	Key use
Radio	km to m	AM/FM radio, TV broadcasting
Microwave	cm to mm	Microwave ovens, mobile phones, radar
Infrared (IR)	μm	TV remotes, thermal cameras, heat sensing
Visible	400–700 nm	Human vision (VIBGYOR)
Ultraviolet (UV)	<400 nm	Sterilisation, vitamin D synthesis, fluorescence
X-rays	0.01–10 nm	Medical imaging, airport security
Gamma	<0.01 nm	Cancer radiotherapy, sterilising medical equipment

PART E — ELECTRICITY AND MAGNETISM

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter E1 — Current Electricity

- **Charge (q)** — unit: coulomb (C).

- **Current (I)** = charge per second = q/t . Unit: ampere (A).
- **Voltage / Potential difference (V)** — the "electrical pressure" that drives current. Unit: volt (V).
- **Resistance (R)** — opposition to current. Unit: ohm (Ω).

Ohm's Law

$$V = I \times R$$

- At constant temperature, current through a conductor is **proportional to voltage**.
- **V** = voltage (volts), **I** = current (amperes), **R** = resistance (ohms).

Power formulas (all equivalent — pick the one matching what you know):

- $P = V \times I$
- $P = I^2 \times R$
- $P = V^2 / R$

Unit: **watt (W)**. Commercial unit: **kilowatt-hour (kWh)** = 1 unit on your bill = 3.6×10^6 J.

Household Electricity in India

230 V, 50 Hz, AC. The 230 V is the RMS voltage. USA uses 120 V, 60 Hz.

- **AC (Alternating Current)** — current reverses direction periodically (50 times per second in India = 50 Hz). Used for transmission over long distances because voltage can be stepped up or down using transformers.
- **DC (Direct Current)** — current flows in one direction only. Used in batteries, electronic devices.

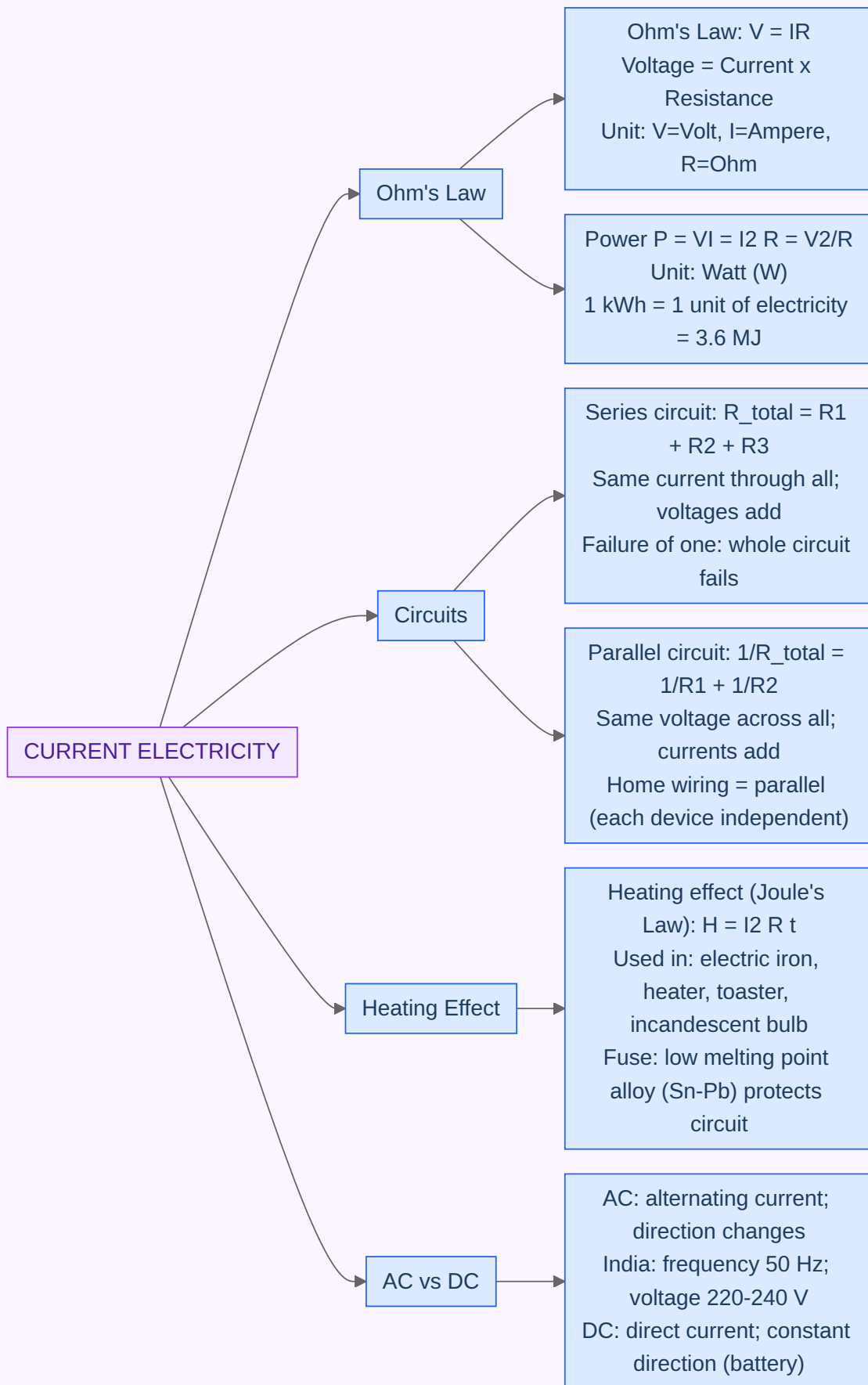
EXAM ALERT

Why is AC preferred for power transmission? Transformers can step up voltage for long-distance transmission (reducing current $\rightarrow$ reducing heat loss in wires: $P = I^2 R$). At the destination, step it back down for safe home use. Transformers work only on AC, not DC — because DC doesn't create a changing magnetic flux, so no induction occurs.

Circuits and Safety

- **Fuse** — a thin wire of low-melting-point alloy that melts when current exceeds a safe limit, breaking the circuit.
- **MCB (Miniature Circuit Breaker)** — modern alternative; trips a switch instead of burning a wire; can be reset.
- **Short circuit** — accidental path of near-zero resistance; causes very high current $\rightarrow$ heat $\rightarrow$ fire.
- **Earthing** — safety connection to the ground. Any leakage current flows harmlessly into the ground instead of through a person.

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter E2 — Magnetism and Electromagnetic Induction

Magnets

- A bar magnet has two poles — North and South. Like poles repel; unlike poles attract.
- Earth is a giant magnet. The geographic North Pole of Earth is near the magnetic South Pole (that's why the North-seeking pole of a compass needle points north — it's attracted to Earth's magnetic south).
- Magnetic field lines run from North to South outside the magnet (and South to North inside).

Electromagnetic Induction — Faraday's Discovery (1831)

Moving a magnet near a coil of wire induces a voltage (EMF) in the coil, and therefore a current if the circuit is complete.

KEY FACT

This is the foundation of everything electrical we use. Generators in power stations, microphones, guitar pickups, wireless charging — all are applications of electromagnetic induction. Michael Faraday made this discovery in 1831, and it changed civilisation.

Lenz's Law

The direction of the induced current is always such that it opposes the change (motion or flux change) causing it. Think of it as electromagnetic braking — nature resists change.

Motor and Generator

Device	Converts	Principle	Hand Rule
Electric motor	Electrical → Mechanical	Current-carrying conductor in a magnetic field experiences a force	Fleming's Left-Hand Rule
Generator / Dynamo	Mechanical → Electrical	Moving conductor in a magnetic field induces current	Fleming's Right-Hand Rule

REMEMBER

Fleming's Left-Hand Rule for Motor: Hold your left hand with thumb, index finger, and middle finger all at right angles to each other. Index finger = direction of magnetic Field, Middle finger = direction of Current, Thumb = direction of Thrust (motion/force on conductor). For a generator, use the Right hand with the same logic — Motion thumb, Field index, Current middle.

Motor = Left = I drive the Left side. Generator = Right.

Quick hook: "FBI for Left (Motor)" — Field (index), Basically in the middle (current = Middle finger), Indicates force/Thrust (thumb). Left hand = Lets electricity Lift things (motor). Right hand = Rotation Releases current (generator).

EXAM ALERT

⚠ TRAP: Motor uses **Left** hand; Generator uses **Right** hand. Students commonly swap them. Memory: moLtor → Left. Generators Rotate to produce current → Right hand. In the exam: if it converts electrical → mechanical = motor = LEFT. Mechanical → electrical = generator = RIGHT.

Transformer

Changes the voltage of AC supply.

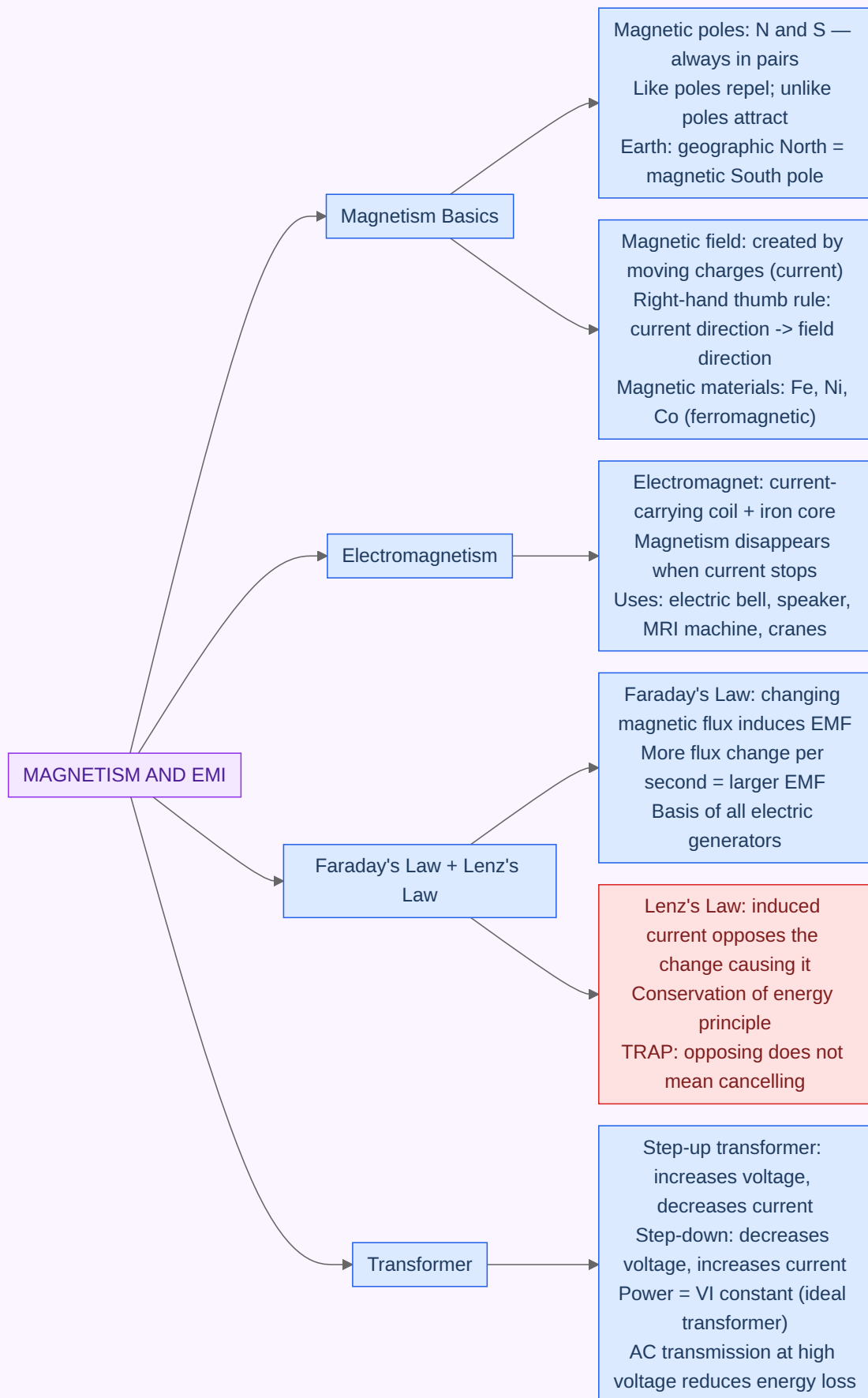
- **Step-up transformer:** increases voltage, decreases current.

- **Step-down transformer:** decreases voltage, increases current.

Works on mutual induction — the changing current in the primary coil creates a changing magnetic field that induces a current in the secondary coil. Transformers work only with AC.

PART F — MODERN PHYSICS

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F1 — Atomic Models

Model	Year	Proposer	Key idea
Plum-pudding	1897	J.J. Thomson	Atom = positive sphere with electrons embedded like plums
Nuclear	1911	Ernest Rutherford	Most of atom is empty space; tiny dense nucleus at centre
Bohr model	1913	Niels Bohr	Electrons orbit in fixed quantised energy levels
Quantum mechanical	1926	Schrödinger / Heisenberg	Electrons exist as probability clouds (orbitals)

Rutherford used the famous gold-foil (alpha-scattering) experiment — firing alpha particles at gold foil. Most passed through (empty space), some deflected slightly (electrons), and a few bounced straight back (nucleus). He concluded the atom is mostly empty with a tiny, dense, positive nucleus.

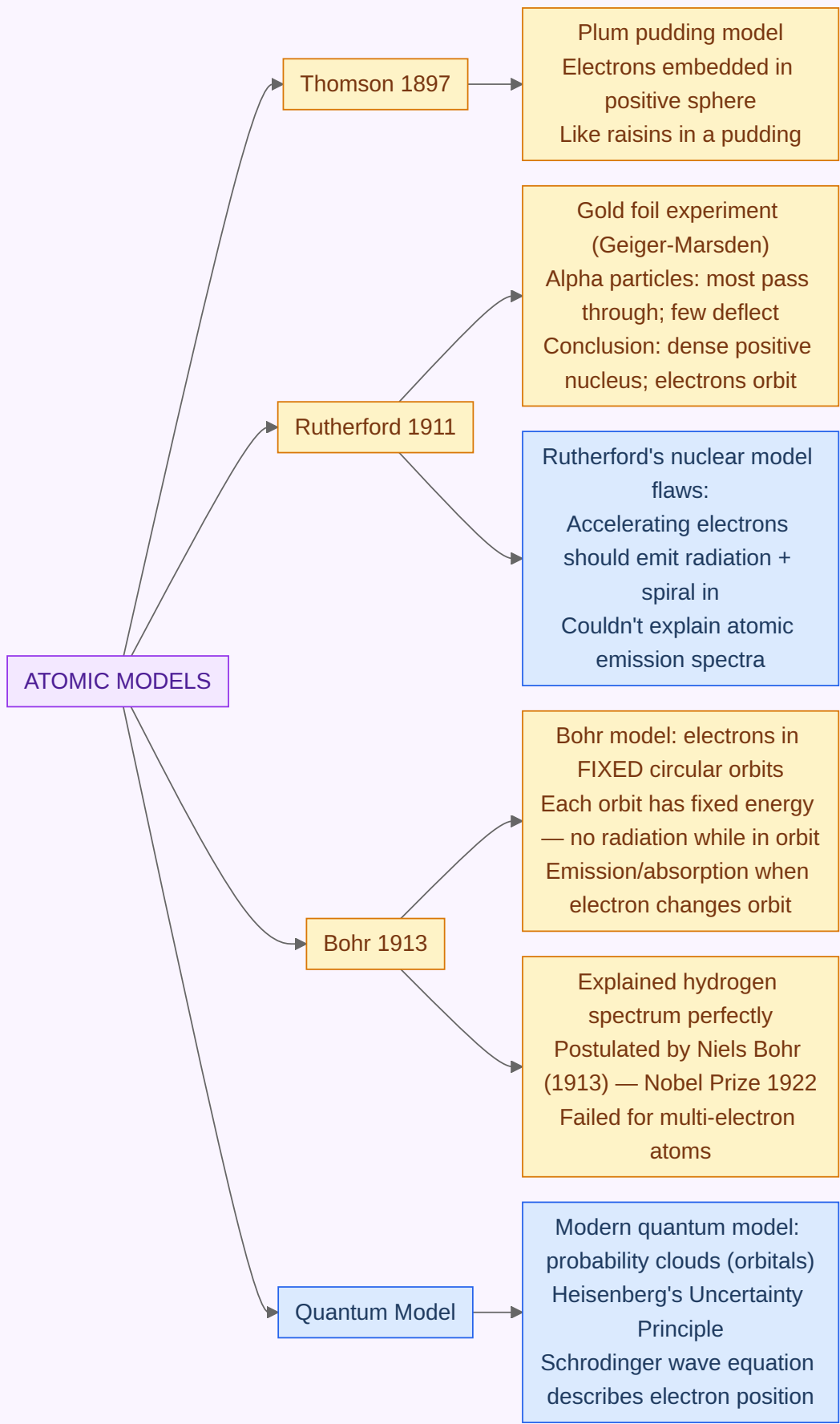
KEY FACT

Bohr model — energy of hydrogen electron in orbit n :

$$E_n = -\frac{13.6}{n^2} \text{ eV}$$

- $n = 1$ (ground state): $E = -13.6 \text{ eV}$ (most stable)
- $n = 2$: $E = -3.4 \text{ eV}$
- $n = 3$: $E = -1.51 \text{ eV}$
- The **negative sign** means the electron is bound to the nucleus; energy needed to free it = 13.6 eV (ionisation energy of hydrogen).
- When an electron **falls** from a higher n to a lower n , it **emits** a photon. When it **absorbs** energy, it **jumps** to a higher n .
- Energy of emitted photon: $E = hf = 13.6 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \text{ eV}$, where $n_2 > n_1$.

CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



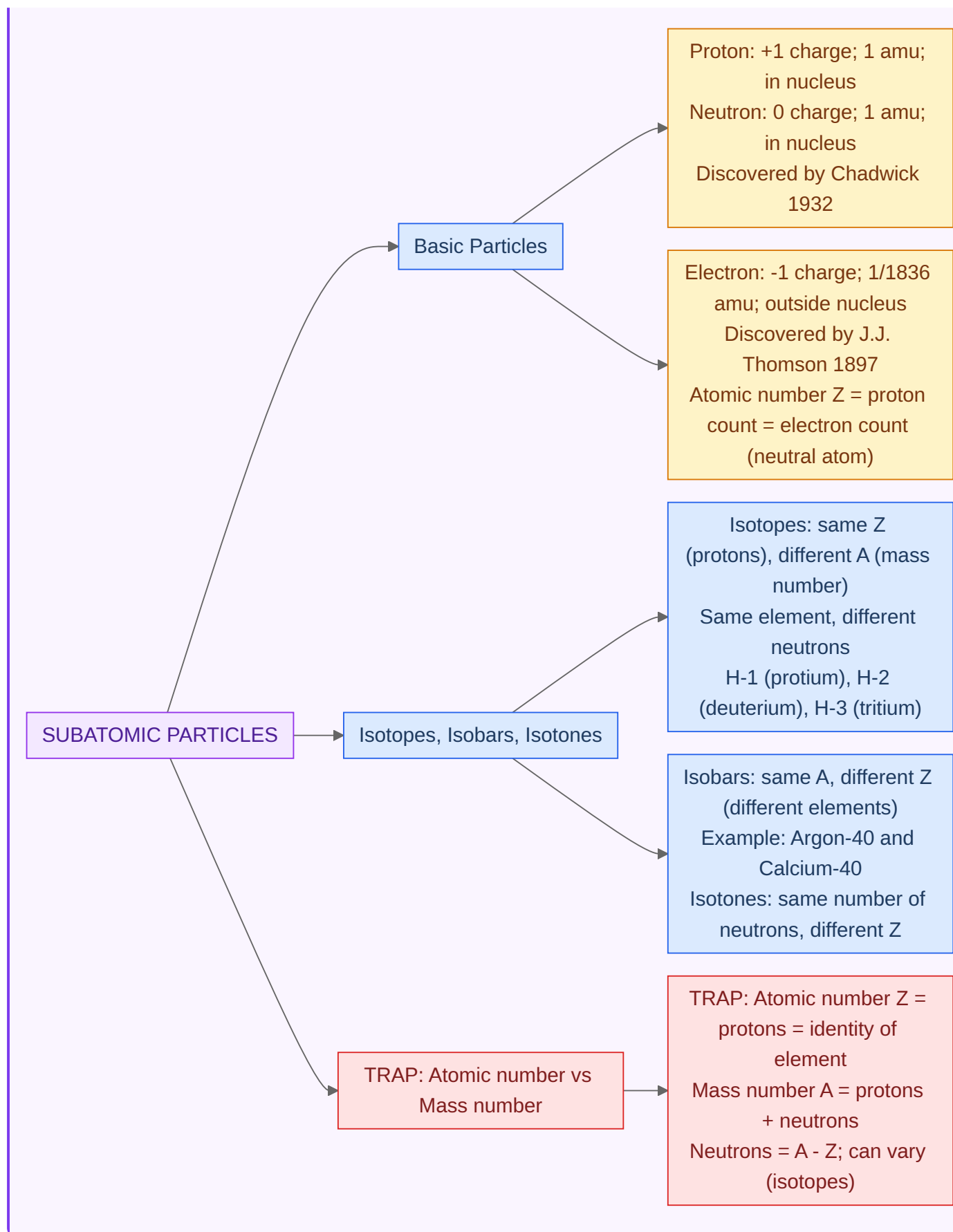
Chapter F2 — Subatomic Particles

Particle	Charge	Relative mass	Discoverer	Year
Electron	-1	1/1837	J.J. Thomson	1897
Proton	+1	1	Rutherford	1919
Neutron	0	~1 (slightly > proton)	James Chadwick	1932

KEY FACT

The heaviest sub-atomic particle is the **neutron** (just slightly heavier than the proton). Three quick-recall facts: electron → Thomson 1897, proton → Rutherford 1919, neutron → Chadwick 1932. The first Nobel Prize in Physics (1901) went to Röntgen for X-rays.

© CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F3 — Radioactivity

Henri Becquerel discovered radioactivity in 1896 (from uranium). Marie and Pierre Curie then isolated polonium (named after Marie's homeland Poland) and radium in 1898. Marie Curie won two Nobel Prizes — Physics 1903 and Chemistry 1911.

Three Types of Radiation

Type	Symbol	Nature	Charge	Penetrating power	Ionizing power	Stopped by
Alpha	α	Helium nucleus ($2p + 2n$)	+2	Lowest ($\alpha < \beta < \gamma$)	Highest ($\alpha > \beta > \gamma$)	Paper, few cm of air
Beta	β	Fast electron (or positron)	-1 (or +1)	Medium	Medium	3 mm aluminium
Gamma	γ	High-energy EM wave (photon)	0	Highest	Lowest	Several cm of lead

KEY FACT

Key inverse relationship: Penetrating power and ionizing power are **opposite** for the three radiation types. - α : highest ionizing power (large charge, slow speed $\rightarrow$ many collisions per cm) but lowest penetration (stopped by paper). - γ : highest penetrating power (no charge, massless photon $\rightarrow$ few collisions) but lowest ionizing power. - This makes α particles the most dangerous **internally** (if inhaled/ingested) and γ rays the most dangerous **externally** (penetrate the body).

REMEMBER

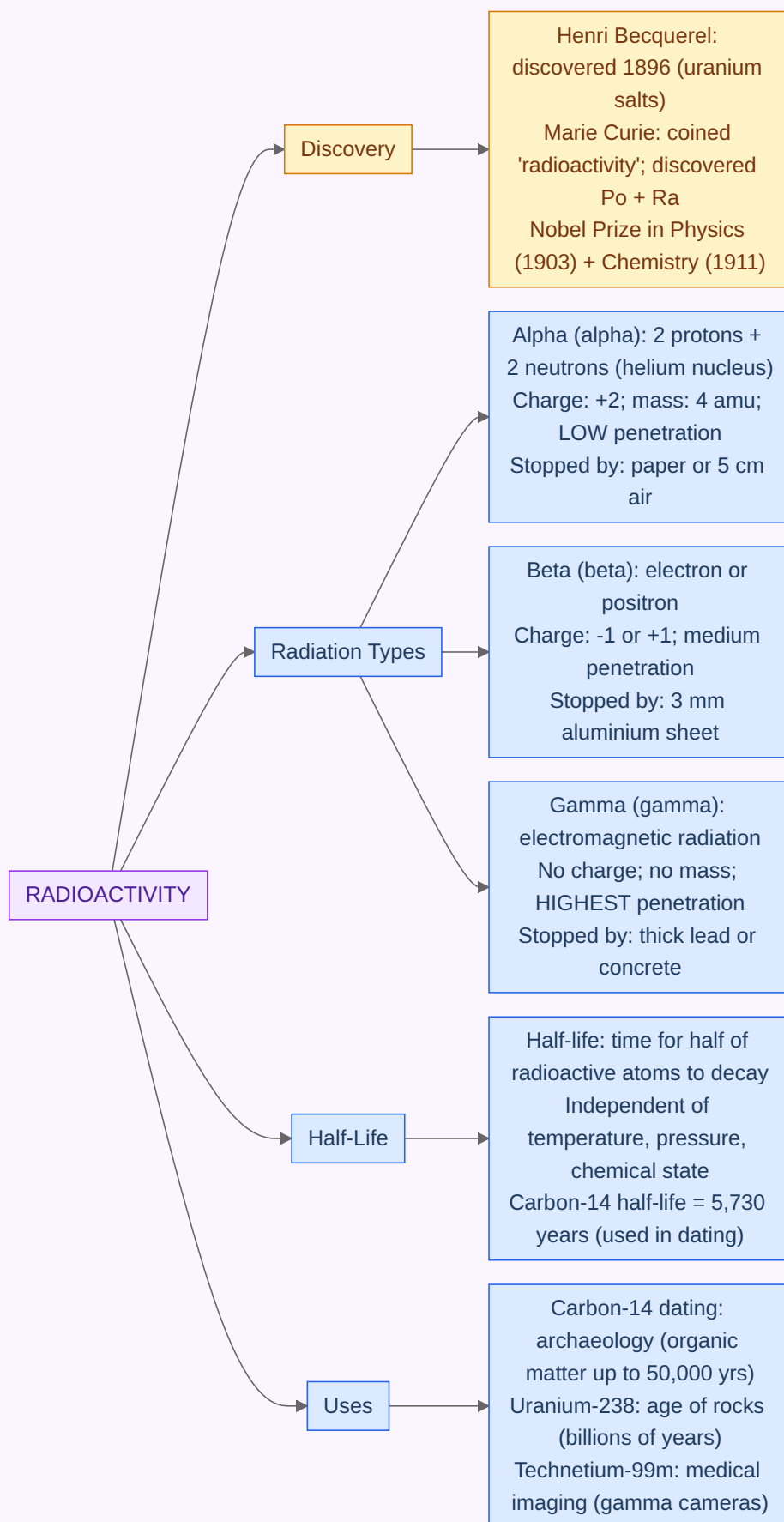
Penetrating power order: $\alpha < \beta < \gamma$ (alphabetical = weakest to strongest). What stops them: **Paper, Aluminium, Lead** — think **PAL**. Alpha needs a PAL to stop it (paper stops alpha). Aluminium stops beta. Lead stops gamma.

Analogy: Alpha is like a heavy truck — massive and easy to stop (paper does it). Beta is like a bicycle — lighter, needs aluminium. Gamma is like a ghost (pure energy, massless) — needs thick lead to absorb it.

EXAM ALERT

⚠ TRAP: "What stops alpha radiation?" — Many students write lead (since lead is the 'strongest' shield). But alpha is the weakest penetrator and is stopped by just a sheet of paper or a few cm of air. Lead is for gamma. Alpha $\rightarrow$ Paper | Beta $\rightarrow$ Aluminium | Gamma $\rightarrow$ Lead (PAL).

🕒 CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F4 — Nuclear Reactions

Fission

A heavy nucleus (Uranium-235 or Plutonium-239) splits into smaller fragments + 2-3 neutrons + enormous energy.

Each released neutron can trigger more fissions → chain reaction. In a nuclear reactor, control rods (cadmium or boron) absorb excess neutrons to control the rate. In an atomic bomb, the chain reaction is uncontrolled.

Fusion

Light nuclei (hydrogen isotopes — deuterium and tritium) combine to form helium + energy.

This is the source of the Sun's energy. Fusion releases far more energy per unit mass than fission, but controlled fusion on Earth has not yet been achieved commercially. The ITER project in France aims to demonstrate controlled fusion.

EXAM ALERT

"Sun's energy comes from nuclear fusion" — this appears in almost every exam. Hydrogen (specifically deuterium and tritium) fuses to form helium, releasing energy according to Einstein's $E = mc^2$. The mass of the helium product is slightly less than the combined mass of the inputs — that tiny mass difference becomes an enormous amount of energy.

REMEMBER

Fission vs Fusion — easy memory split: "Fission = Fight" (heavy nucleus breaks apart, like splitting a fighting duo). "Fusion = Funnel" (light nuclei come together and merge). Fission = atom bomb (U-235). Fusion = hydrogen bomb and the SUN. Fusion releases more energy per kg but is harder to control.

EXAM ALERT

⚠ TRAP: "Sun's energy source?" — Answer is nuclear **fusion** (NOT fission). Many students write fission because nuclear reactors use fission. The Sun fuses hydrogen into helium. All stars are fusion reactors. Commercial electricity uses fission. Fusion is what powers stars; fission is what powers current reactors.

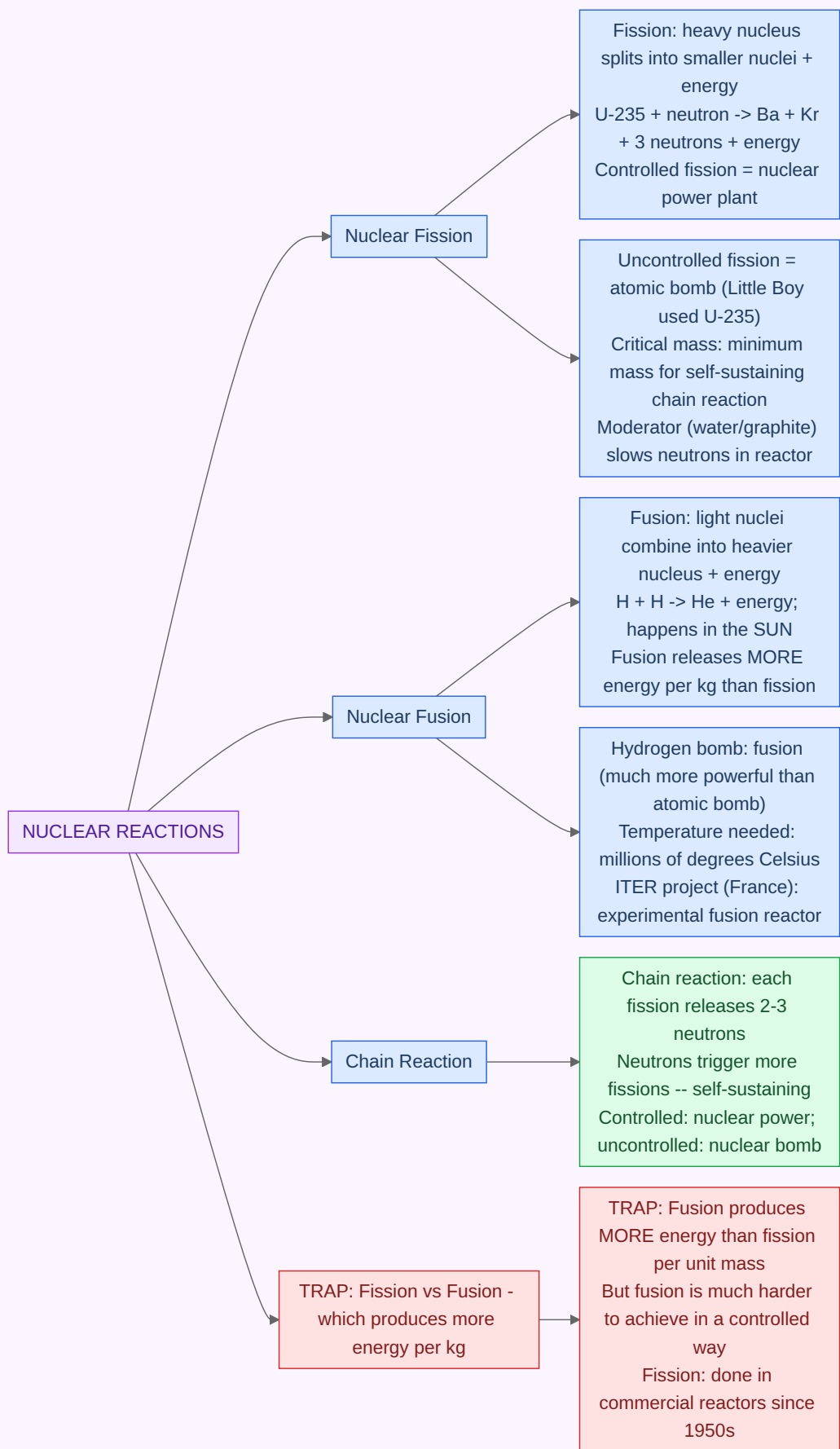
$E = mc^2$

Einstein's mass-energy equivalence. Even a tiny mass (m) holds an enormous amount of energy ($c = 3 \times 10^8$ m/s, so $c^2 = 9 \times 10^{16}$). Burning 1 gram of matter completely would yield 9×10^{13} joules — equivalent to 20,000 tons of TNT.

Radiocarbon Dating

Carbon-14 (^{14}C) is a radioactive isotope of carbon with half-life of **5730 years**. Living organisms maintain a constant $^{14}\text{C}/^{12}\text{C}$ ratio by continuously taking in carbon. When they die, ^{14}C decays without being replenished. Measuring the remaining ^{14}C ratio tells us the age of the organism. Used in archaeology (up to ~50,000 years back).

◎ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Chapter F5 — Semiconductors

F5.1 — Classification of Solids by Conductivity

Type	Resistivity	Free carriers	Examples
Conductor (metal)	$< 10^{-8} \Omega \cdot \text{m}$	Free electrons (many)	Copper, silver, aluminium
Semiconductor	10^{-5} to $10^3 \Omega \cdot \text{m}$	Electrons + holes (few)	Silicon (Si), Germanium (Ge)
Insulator	$> 10^8 \Omega \cdot \text{m}$	None	Glass, rubber, plastic

KEY FACT

Key difference from metals: Metal resistance **increases** with temperature (electrons collide more). Semiconductor resistance **decreases** with temperature — heat frees more charge carriers. This is the opposite behaviour, and it's a classic exam question.

Which semiconductor is more common? Silicon (Si) — it is cheaper, widely available (sand is SiO_2), and remains stable at higher temperatures. Germanium (Ge) was used in early transistors but largely replaced by Si.

Both Si and Ge are in **Group 14** of the periodic table (4 valence electrons each).

F5.2 — Intrinsic vs Extrinsic Semiconductors

Type	What it is	Majority charge carriers
Intrinsic	Pure Si or Ge crystal; no dopant	Electrons = holes (equal numbers)
Extrinsic	Si or Ge doped with impurity atoms	Unequal — depends on dopant
n-type	Pentavalent dopant added (P, As, Sb — Group 15)	Electrons are majority carriers
p-type	Trivalent dopant added (B, Al, Ga, In — Group 13)	Holes are majority carriers

Doping = deliberate addition of tiny amounts of impurity atoms to dramatically increase conductivity.

- **n-type:** Phosphorus has 5 valence electrons. 4 bond with Si neighbours; the 5th is loosely held and becomes a free electron. Net charge of the crystal is still neutral, but electrons are the majority carriers.
- **p-type:** Boron has 3 valence electrons. It forms bonds with 3 Si neighbours but leaves one bond incomplete — this "missing electron" is called a **hole**. Holes act as positive charge carriers.

COMMON TRAP

⚠ TRAP: "What are majority carriers in p-type?" — Answer: **Holes** (NOT electrons). In p-type, holes are majority and electrons are minority carriers. Students often say "positive charges" and while conceptually right, the precise answer is "holes." Similarly, in n-type, electrons are majority carriers and holes are minority carriers.

F5.3 — p-n Junction Diode

A p-n junction is formed by joining a p-type and n-type semiconductor in a single crystal.

- **Depletion layer:** At the junction, electrons from n-side diffuse into p-side and fill holes. This creates a region with no free carriers and a built-in electric field pointing from n to p.
- **Forward bias:** p-side connected to +ve terminal of battery; n-side to -ve. The external field opposes the built-in field → depletion layer narrows → current flows easily.

- **Reverse bias:** p-side connected to -ve terminal. External field reinforces built-in field → depletion layer widens → current blocked (only a tiny leakage current flows).

KEY FACT

Key application — Rectification: A diode converts AC (which alternates direction) to DC (one direction only) because it only allows current in one direction. A **half-wave rectifier** uses one diode; a **full-wave rectifier** uses four diodes in a bridge configuration.

Zener diode: A specially doped diode designed to operate in reverse bias at a precise "Zener voltage." Used for **voltage regulation** — keeps output voltage constant even if input fluctuates. Unlike a normal diode, reverse breakdown in a Zener is controlled and non-destructive.

F5.4 — Special Diodes and Applications

Device	Operating principle	Primary use
Rectifier diode	Conducts in forward bias only	AC → DC conversion
Zener diode	Controlled reverse-bias breakdown	Voltage regulation / voltage reference
LED (Light Emitting Diode)	Electron-hole recombination emits a photon	Indicator lights, displays, TV backlights
Photodiode	Incident photons generate electron-hole pairs	Light sensors, optical communication
Solar cell	Photovoltaic effect in p-n junction	Sunlight → electricity

LED colour depends on semiconductor material: - GaAs (Gallium Arsenide) → infrared - GaP (Gallium Phosphide) → green / red - GaN (Gallium Nitride) → blue / white (enables white LEDs)

KEY FACT

LED vs Solar cell — easy memory: LED = electricity IN, light OUT (forward biased). Solar cell = light IN, electricity OUT. They are conceptually reverse processes in a p-n junction.

F5.5 — Transistor

A transistor is a three-layer semiconductor device: either **n-p-n** or **p-n-p** sandwich. It has three terminals: - **E** — **Emitter:** supplies majority carriers - **B** — **Base:** thin, lightly doped middle layer that controls current - **C** — **Collector:** collects majority carriers

Two key functions:

1. **Amplifier** — a small base current controls a large collector current. Current gain $\beta = I_C / I_B$ (typically 20 to 500). One milliamp at the base controls hundreds of milliamps at the collector.
2. **Switch** — small base signal turns transistor fully ON (saturated) or fully OFF (cut-off); used in digital logic and computers.

KEY FACT

Transistor was invented in **1947** by John Bardeen, Walter Brattain, and William Shockley at Bell Labs, USA — Nobel Prize in Physics 1956. It replaced vacuum tubes (thermionic valves) making electronics smaller, cheaper, and more reliable. Without the transistor there would be no modern computers, smartphones, or satellites.

NPN transistors are more common than PNP because electrons (majority carriers in NPN) have higher mobility than holes.

F5.6 — Logic Gates (Digital Electronics)

Logic gates are the building blocks of all digital circuits (computers, phones, calculators). They take binary inputs (0 or 1) and produce a binary output.

Gate	Boolean expression	Output rule	Truth table (key cases)
AND	$A \cdot B$	Output = 1 only if both inputs are 1	$1 \cdot 1 = 1$; all others = 0
OR	$A + B$	Output = 1 if any input is 1	$0 + 0 = 0$; all others = 1
NOT	$\bar{A}$ (inverter)	Output = opposite of input	$0 \rightarrow 1$; $1 \rightarrow 0$
NAND	$\text{NOT}(A \cdot B)$	Output = 0 only if both inputs are 1	opposite of AND
NOR	$\text{NOT}(A + B)$	Output = 1 only if both inputs are 0	opposite of OR
XOR	$A \oplus B$	Output = 1 if inputs differ	$0, 0 \rightarrow 0$; $0, 1 \rightarrow 1$; $1, 0 \rightarrow 1$; $1, 1 \rightarrow 0$

KEY FACT

Universal gates: NAND and NOR are called universal gates because **any** logic gate (AND, OR, NOT, XOR...) can be built using only NAND gates alone, or only NOR gates alone. This is very important in chip manufacturing — having one type of gate simplifies fabrication.

COMMON TRAP

⚠ Classic exam traps on logic gates:

- "Which gate gives output 1 when both inputs are 0?" → **NOR gate** (not OR, not AND).
- "Which gate gives output 0 when both inputs are 1?" → **NAND gate** (not AND).
- "Universal gates?" → **NAND and NOR** (not AND, not OR).
- "XOR gives 1 when inputs are ___?" → **Different** (0,1 or 1,0). Same inputs (0,0 or 1,1) give 0.
- "NOT gate is also called?" → **Inverter**.

F5.7 — Integrated Circuits (ICs) and Moore's Law

An **Integrated Circuit (IC)** places thousands to billions of transistors, resistors, and capacitors on a single tiny silicon chip (a few mm^2).

Key facts: - Invented by **Jack Kilby** (Texas Instruments, 1958) — Nobel Prize in Physics 2000 - Independently co-invented by **Robert Noyce** (Fairchild Semiconductor, 1958) — the Nobel went to Kilby alone (Noyce died in 1990) - **Generations of ICs** (by transistor count): - SSI (Small Scale Integration) — tens of transistors - MSI (Medium Scale Integration) — hundreds - LSI (Large Scale Integration) — thousands - VLSI (Very Large Scale Integration) — millions (1980s–1990s) - ULSI (Ultra Large Scale Integration) — billions (modern CPUs, GPUs) - **Moore's Law** (Gordon Moore, 1965): the number of transistors on a chip doubles approximately every 2 years. This has held roughly true for six decades.

Semiconductor Trap Card — five facts that repeatedly appear in SSC CGL/NTPC papers:

1. "Which semiconductor is more commonly used — Si or Ge?" → **Silicon (Si)** — stable at higher temperatures, cheaper, more abundant.
2. "Majority carriers in p-type?" → **Holes** (NOT electrons; electrons are minority carriers in p-type).
3. "Universal gates?" → **NAND and NOR** (any circuit can be built from NAND alone, or NOR alone).
4. "LED works on which bias?" → **Forward bias** (it emits light when forward biased).
5. "Who invented the IC?" → **Jack Kilby** (Nobel Prize 2000). Robert Noyce co-invented independently the same year but the Nobel went to Kilby alone.

PART V — INSTRUMENTS AND WHAT THEY MEASURE

Instrument	Measures
Ammeter	Electric current (connected in series)
Voltmeter	Potential difference / voltage (connected in parallel)
Galvanometer	Very small electric currents
Multimeter	Voltage, current, and resistance
Wattmeter	Electric power
Energy meter	Electrical energy consumed (kWh)
Ohmmeter	Electrical resistance
Thermometer	Temperature
Pyrometer	Very high temperatures (furnaces, molten metal)
Calorimeter	Heat exchanged in reactions
Barometer	Atmospheric pressure
Manometer	Gas pressure
Hygrometer / Psychrometer	Relative humidity
Anemometer	Wind speed
Rain gauge (Pluviometer)	Rainfall
Seismograph	Earthquake intensity and location
Lactometer	Purity / density of milk
Hydrometer	Density of liquids
Saccharometer	Sugar content of a solution
Salinometer	Salt concentration in water
Speedometer	Speed of a vehicle
Odometer	Distance travelled by a vehicle
Tachometer	Rotational speed (RPM) of a shaft or engine
Sextant	Angular altitude of celestial bodies (navigation)
Theodolite	Horizontal and vertical angles (land surveying)
Audiometer	Hearing sensitivity (threshold of hearing)
Stethoscope	Heart and lung sounds
Sphygmomanometer	Blood pressure
Cardiograph (ECG)	Electrical activity of the heart
Electroencephalograph (EEG)	Electrical activity of the brain
Tonometer	Eye pressure (glaucoma screening)
Endoscope	Images from inside the body
Periscope	Viewing over obstacles (used in submarines)
Spectrometer	Wavelengths of light / analysis of light spectrum
Refractometer	Refractive index of a substance
Photometer	Intensity of light
Bolometer	Power of incident electromagnetic radiation
Oscilloscope	Waveform of electrical signals

PART W — DISCOVERIES AND INVENTIONS IN PHYSICS

Discovery / Invention	Scientist	Year	Country
Compound microscope	Zacharias Janssen	1590	Netherlands
Telescope (refractor)	Galileo Galilei	1609	Italy
Reflecting telescope	Isaac Newton	1668	England
Laws of motion, universal gravitation	Isaac Newton	1687	England
Electrostatic machine (Wimshurst)	Robert Van de Graaff	1929	USA
Barometer	Evangelista Torricelli	1643	Italy
Steam engine (improved)	James Watt	1769	Scotland
Battery (voltaic pile)	Alessandro Volta	1800	Italy
Electromagnetic induction	Michael Faraday	1831	England
Laws of thermodynamics (conservation of energy)	Helmholtz, Joule, Mayer	1840s	Germany, England
Telephone	Alexander Graham Bell	1876	Scotland/USA
Electric light bulb (incandescent)	Thomas Edison	1879	USA
Wireless (radio) transmission	Guglielmo Marconi	1895	Italy
Wireless waves (millimetre waves)	Jagadish Chandra Bose	1895	India
X-rays	Wilhelm Röntgen	1895	Germany
Radioactivity	Henri Becquerel	1896	France
Polonium + Radium	Marie and Pierre Curie	1898	Poland/France
Quantum hypothesis	Max Planck	1900	Germany
Aeroplane (first flight)	Wright Brothers	1903	USA
Special Theory of Relativity, photoelectric effect	Albert Einstein	1905	Germany
Bohr atomic model	Niels Bohr	1913	Denmark
Television (mechanical)	John Logie Baird	1925	Scotland
Raman Effect	C.V. Raman	1928	India
Cyclotron (particle accelerator)	Ernest Lawrence	1929	USA
Nuclear chain reaction	Enrico Fermi	1942	Italy/USA
Transistor	Bardeen, Brattain, Shockley	1947	USA
Holography	Dennis Gabor	1948	Hungary/UK
Laser	Theodore Maiman	1960	USA
Integrated circuit	Jack Kilby	1958	USA
Optical fibre communication	Charles Kao	1966	China/UK
Gravitational waves detected (LIGO)	LIGO team (Thorne, Weiss, Barish)	2015/2017	USA
Higgs boson confirmed	CERN experiments	2012	International
Bose-Einstein statistics	S.N. Bose + Einstein	1924	India/Germany
Chandrasekhar limit	S. Chandrasekhar	1930	India/USA

EXAM ALERT

Indian scientists to remember for physics: **C.V. Raman** (Raman Effect, Nobel Prize in Physics 1930 — the only Nobel in Physics won by an Indian as of 2026), **Jagadish Chandra Bose** (millimetre/microwave waves, 1895 — before Marconi in some interpretations), **Satyendra Nath Bose** (Bose-Einstein statistics,

1924 — the "boson" in Higgs boson is named after him), **Subrahmanyan Chandrasekhar** (Chandrasekhar limit, Nobel 1983).

PART X — DAILY-LIFE PHYSICS ("WHY" QUESTIONS)

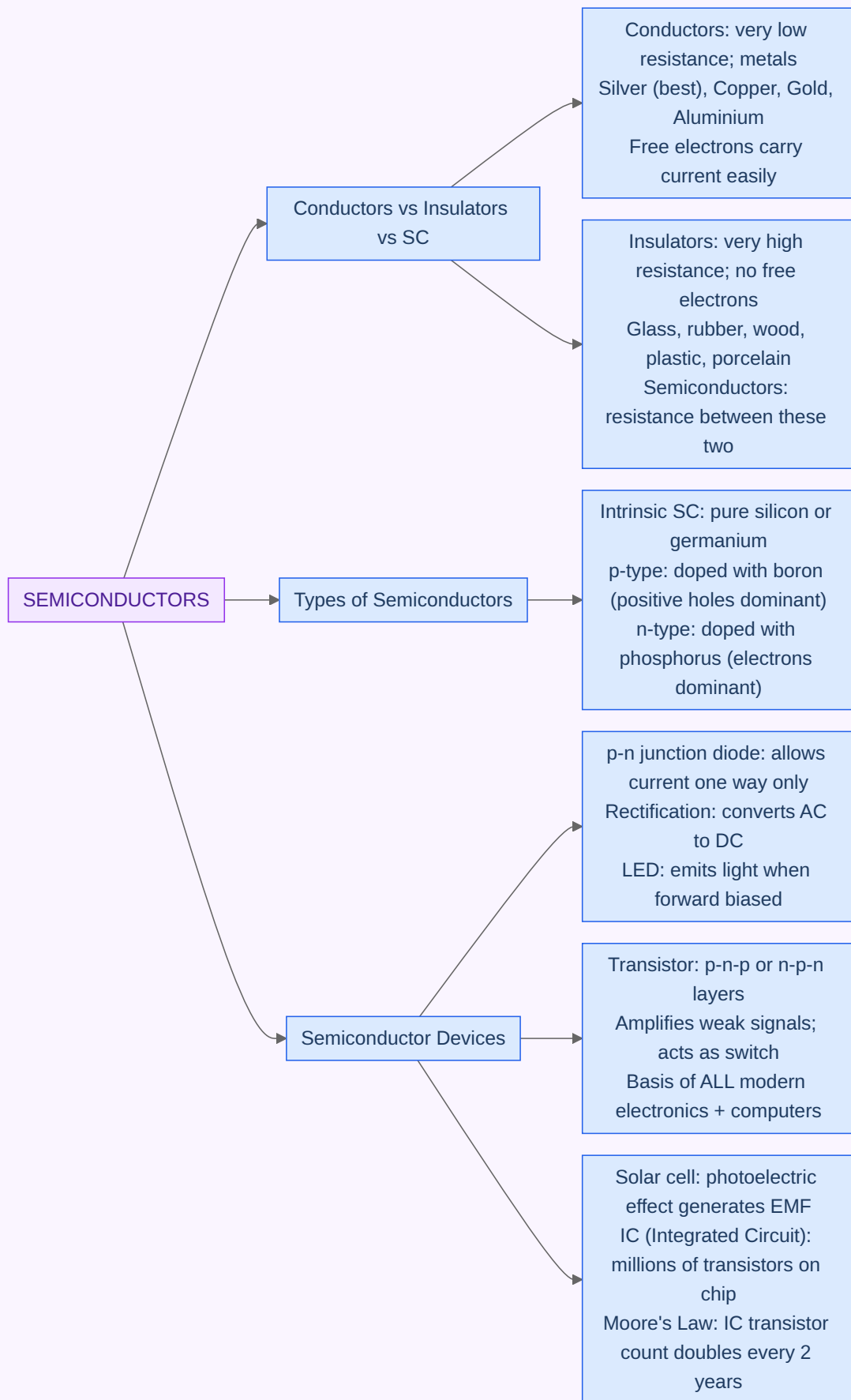
1–2 questions on everyday physics phenomena appear in almost every competitive exam. For each phenomenon below, know both the fact and the reason behind it.

Phenomenon	Explanation
Pressure cooker cooks faster	Higher pressure raises the boiling point of water to $\sim 120^\circ\text{C}$; food cooks faster at higher temperature
Cooking takes longer on mountains	Atmospheric pressure is lower at high altitude; water boils at lower temperature ($\sim 70^\circ\text{C}$ on Everest); food cooks slower
Ice floats on water	Anomalous expansion — water expands on freezing; ice density (0.92 g/cm^3) < water density (1.0 g/cm^3)
Fish survive under frozen lake	Ice forms on top and insulates the water below; liquid water at $\sim 4^\circ\text{C}$ stays at the bottom where fish live
Mirage in desert	Light bends due to temperature gradient in hot air near the ground — total internal reflection of sky light
Stars twinkle, planets don't	Stars are point sources — atmospheric turbulence causes their single ray to flicker. Planets are extended discs — their many rays average out, no twinkling
Sky is blue	Rayleigh scattering — shorter (blue) wavelengths scatter much more than longer (red) wavelengths
Sunset and sunrise are red/orange	Sunlight travels a long path through the atmosphere; blue has already scattered away; red, which scatters least, survives to reach your eyes
Rainbow forms after rain	Water droplets act as tiny prisms — refraction + total internal reflection + dispersion separate white light into VIBGYOR
Diamond sparkles	Very high refractive index (2.42) $\rightarrow$ small critical angle ($\sim 24^\circ$) $\rightarrow$ total internal reflection of most rays $\rightarrow$ sparkle
Soap film shows colours	Thin-film interference — light reflects from both surfaces of the thin film; they interfere constructively/destructively at different wavelengths
Why we sweat to cool	Evaporation of sweat absorbs latent heat of vaporisation from the skin surface
Steam burn worse than water burn	Steam at 100°C releases latent heat of vaporisation (2260 J/g) when it condenses on skin, plus heat from cooling
We slip on ice	Pressure of body weight on ice slightly lowers its melting point; a thin film of water forms $\rightarrow$ low friction
Spoon appears bent in water	Refraction — light bends when crossing from water to air; the image appears to shift
Lightning seen before thunder	Light travels at $3 \times 10^8\text{ m/s}$, sound at $\sim 340\text{ m/s}$; light arrives almost instantly, sound takes ~ 3 seconds per km
Ceiling fan cools us	Increases evaporation of sweat from skin; does not actually cool the room air
Hot tea in saucer cools faster	Larger surface area increases rate of evaporation and radiation
Pendulum clock loses time in summer	Pendulum rod expands with heat $\rightarrow$ longer pendulum $\rightarrow$ longer period $\rightarrow$ clock runs slow
Birds sit on high-tension wires safely	Both feet are on the same wire at the same potential; no potential difference between them $\rightarrow$ no current through bird
A 3-pin plug is safer	Earth pin diverts any leakage current to the ground instead of through the user
Microwave heats food	Microwaves make water molecules vibrate (dielectric heating); heating from inside out
Helicopter needs a tail rotor	Main rotor rotation creates torque on the body (Newton's 3rd Law); tail rotor counters this torque
Geostationary satellite stays above same spot	Orbital period is 24 hours, same as Earth's rotation; appears stationary from Earth
Satellites feel weightless	Satellite and astronaut are in free fall together; apparent weight = 0 (no contact force needed)

Phenomenon	Explanation
Echo minimum distance is 17 m	Human ear needs 0.1 s gap to distinguish echo from original sound; at 340 m/s, minimum round trip = 34 m → 17 m one way
Mountaineers carry oxygen cylinders	Atmospheric pressure at high altitude is very low; partial pressure of oxygen is insufficient for breathing

PART AA — KEY CONSTANTS AND FORMULAS

⦿ CHAPTER SUMMARY — EVERYTHING IN ONE VIEW



Physical Constants

Constant	Symbol	Value
Speed of light in vacuum	c	3×10^8 m/s
Acceleration due to gravity (Earth surface)	g	9.8 m/s ²
Universal gravitational constant	G	6.674×10^{-11} N·m ² /kg ²
Planck's constant	h	6.626×10^{-34} J·s
Charge of electron	e	1.6×10^{-19} C
Mass of electron	m _e	9.11×10^{-31} kg
Mass of proton	m _p	1.67×10^{-27} kg
Avogadro's number	N _a	6.022×10^{23}
Gas constant	R	8.314 J/mol/K
Boltzmann constant	k	1.38×10^{-23} J/K
Stefan-Boltzmann constant	σ	5.67×10^{-8} W/m ² /K ⁴
Atmospheric pressure	—	101,325 Pa ≈ 760 mm Hg

Key Non-SI Conversions

- 1 horsepower (HP) = 746 watts
- 1 kWh (1 unit of electricity) = 3.6×10^6 joules
- 1 calorie = 4.184 joules
- 1 eV (electron volt) = 1.6×10^{-19} J
- 1 light-year = 9.46×10^{15} metres
- 1 ångström = 10^{-10} metres
- 0 °C = 273.15 K = 32 °F

Important Numerical Values to Memorise

- Escape velocity from Earth = **11.2 km/s**
- Orbital velocity at Earth surface = **7.9 km/s**
- Geostationary orbit altitude = **36,000 km**
- Speed of sound in air (20 °C) = **343 m/s**
- Specific heat of water = **4184 J/kg/°C**
- Latent heat of fusion of ice = **334 J/g**
- Latent heat of vaporisation of water = **2260 J/g**
- India household supply = **230 V, 50 Hz AC**
- Refractive index of diamond = **2.42**
- Refractive index of water = **1.33**

Formula Quick Reference

Mechanics: $F = ma$ | $W = mgh$ | $KE = \frac{1}{2}mv^2$ | $P = W/t$ | $v^2 = u^2 + 2as$

Heat: $Q = mc\Delta T$ | $Q = mL$ | $F = (9/5)C + 32$ | $K = C + 273$

Light: $n = c/v_{\text{medium}}$ | $\sin(\theta_c) = 1/n$ | $1/v - 1/u = 1/f$ | $P = 1/f$ (diopetre)

Electricity: $V = IR$ | $P = VI = I^2R$ | $E = Pt$ | series: $R = R_1 + R_2$ | parallel: $1/R = 1/R_1 + 1/R_2$

Modern: $E = mc^2$ | $E = hf$ | $\lambda = h/p$ (de Broglie)

APPENDIX A — 25-Question Mini-Mock

Attempt all 25 without looking at answers. Time yourself — you should finish in 20 minutes.

1. What is the SI unit of pressure?
2. Who discovered electromagnetic induction and in what year?
3. Escape velocity from Earth is approximately how much?
4. Why does a pressure cooker cook food faster than an open pot?
5. Which type of lens corrects myopia?
6. Speed of sound in air at 0 °C?
7. Which type of mirror is used as a rear-view mirror in vehicles and why?
8. What is the minimum distance needed to hear an echo?
9. State Fleming's Left-Hand Rule (what it applies to).
10. What is 1 horsepower in watts?
11. What radiation has the highest penetrating power — alpha, beta, or gamma?
12. What is the source of the Sun's energy?
13. Who discovered X-rays and in what year?
14. Name three applications of total internal reflection.
15. The sky is blue due to which type of scattering?
16. A transformer works on which principle and which type of current?
17. Which scientist first explained the photoelectric effect?
18. Latent heat of vaporisation of water is approximately how much?
19. What is the commercial unit of electrical energy?
20. Why do stars twinkle but planets do not?
21. Which law of thermodynamics says entropy always increases?
22. Name the person who discovered the neutron.
23. What does SONAR stand for and what does it use?
24. Refractive index of diamond is approximately how much?
25. Indian household electricity supply is at what voltage and frequency?

Answers: 1. Pascal (Pa) 2. Michael Faraday, 1831 3. 11.2 km/s 4. Higher pressure raises the boiling point of water to ~120 °C; food cooks faster 5. Concave (diverging) lens 6. 331 m/s 7. Convex mirror — because it always gives a virtual, erect, diminished image with a wider field of view 8. 17 metres 9. Applies to an electric motor — Index = Field direction, Middle = Current direction, Thumb = Thrust/motion direction 10. 746 watts 11. Gamma radiation 12. Nuclear fusion (hydrogen nuclei fuse to form helium) 13. Wilhelm Röntgen, 1895 14. Optical fibres, diamonds (sparkle), mirages in desert, periscope 15. Rayleigh scattering 16. Mutual induction; AC (Alternating Current) only 17. Albert Einstein (Nobel Prize 1921) 18. 2260 J/g (at 100 °C) 19. Kilowatt-hour (kWh) — also called 1 unit 20. Stars are point sources — atmospheric turbulence flickers their single ray. Planets are extended discs — their multiple rays average out 21. Second Law of Thermodynamics 22. James Chadwick (1932) 23. Sound Navigation and Ranging; uses ultrasonic waves 24. 2.42 25. 230 V, 50 Hz AC

APPENDIX B — Trap Recognition Card

Trap	Common wrong answer	Correct answer
First Nobel Prize in Physics	Einstein, Curie	Röntgen (1901, for X-rays)
Nobel winner for photoelectric effect	J.J. Thomson	Albert Einstein (1921)
Raman Effect Nobel	S.N. Bose	C.V. Raman (1930)
Radio invention — Indian angle	Marconi alone	J.C. Bose also demonstrated millimetre waves in 1895
Sun's energy	Nuclear fission	Nuclear fusion
Most penetrating radiation	Beta	Gamma
What stops alpha radiation	Lead	Paper or a few cm of air
Rear-view mirror type	Concave	Convex (wider field)
Myopia correction	Convex	Concave lens
Speed of sound vs speed of light	Similar	Light is ~880,000× faster
Echo minimum distance	34 m	17 m (half the round trip)
Horsepower in watts	750 W	746 W
Ice floats because...	It is lighter	Anomalous expansion — water expands on freezing
Geostationary satellite height	3600 km	~36,000 km
Transformer works on DC?	Yes	No — only AC creates changing flux